microCT vs Histology: Reproductive Trait Comparison

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1. LOAD PACKAGES

```
library(janitor)      # column name cleaning
library(dplyr)        # data manipulation
library(tidyr)        # data reshaping
library(ggplot2)      # visualisation
library(lme4)         # mixed-effects models
library(lmerTest)     # p-values for lmer
library(emmeans)      # estimated marginal means
library(performance)  # ICC
library(see)      # for model check
```

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2. LOAD AND CLEAN DATA

```
# Read raw data and clean column names
data <- read.csv("microCT_measurements_all.csv", header = TRUE) %>%
  clean_names()

# Quick inspection of dataset structure
str(data)
```

```
## 'data.frame':    2073 obs. of  17 variables:
##  $ specimen_id                   : chr  "P0209" "P0209" "P0209" "P0209" ...
##  $ species_replicate             : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ phylum                        : chr  "Porifera" "Porifera" "Porifera" "Porifera" ...
##  $ species                       : chr  "Monanchora pulchra" "Monanchora pulchra" "Monanchora pulchra
##  $ stage                         : chr  "late oocyte or early embryo" "late oocyte or early embryo"
##  $ type                          : chr  "Gamete" "Gamete" "Gamete" "Gamete" ...
##  $ measurement                   : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ volume                        : num  0.0893 0.2549 0.2585 0.3067 0.3222 ...
##  $ surface_area_lindblad_2005_mm2: num  2.04 2.64 2.98 4.27 2.58 ...
##  $ aspect_ratio                  : num  2.09 2.97 1.4 2.09 1.42 ...
##  $ feret_min_mm                  : num  0.373 0.606 0.721 0.731 0.739 ...
```

```
##  $ feret_mean_mm                : num  0.611 0.88 0.836 0.934 0.922 ...
##  $ feret_max_mm                 : num  0.923 1.31 1 1.26 1.047 ...
##  $ feret_micron                 : num  923 1310 1000 1260 1047 ...
##  $ notes                        : chr  "dense, no clear nucleus, no structure. There is a tiny bit m
##  $ annotation_method            : chr  "Segmentation" "Segmentation" "Segmentation" "Segmentation"
##  $ method                       : chr  "microCT" "microCT" "microCT" "microCT" ...
```

```
head(data)
```

```
##   specimen_id species_replicate   phylum            species
## 1       P0209                 1 Porifera Monanchora pulchra
## 2       P0209                 1 Porifera Monanchora pulchra
## 3       P0209                 1 Porifera Monanchora pulchra
## 4       P0209                 1 Porifera Monanchora pulchra
## 5       P0209                 1 Porifera Monanchora pulchra
## 6       P0209                 1 Porifera Monanchora pulchra
##                          stage   type measurement     volume
## 1 late oocyte or early embryo Gamete           1 0.08928042
## 2 late oocyte or early embryo Gamete           2 0.25489408
## 3 late oocyte or early embryo Gamete           3 0.25850908
## 4 late oocyte or early embryo Gamete           4 0.30666682
## 5 late oocyte or early embryo Gamete           5 0.32216112
## 6 late oocyte or early embryo Gamete           6 0.33206717
##   surface_area_lindblad_2005_mm2 aspect_ratio feret_min_mm feret_mean_mm
## 1                       2.040918     2.089400    0.3733344     0.6113950
## 2                       2.638869     2.973402    0.6063495     0.8803933
## 3                       2.981053     1.400202    0.7205604     0.8361193
## 4                       4.273898     2.086539    0.7306298     0.9343422
## 5                       2.582771     1.421635    0.7388691     0.9215980
## 6                       2.895310     1.513712    0.7475821     1.1014550
##   feret_max_mm feret_micron
## 1    0.9227679     922.7679
## 2    1.3098306    1309.8306
## 3    1.0000609    1000.0609
## 4    1.2597934    1259.7934
## 5    1.0472515    1047.2515
## 6    1.4217539    1421.7539
##
## 1                     dense, no clear nucleus, no structure. There is a tiny bit missing in the segm
## 2 dense, no clear nucleus, no structure. Elongated, on edge.  There is a tiny bit missing in the segm
## 3 dense, no clear nucleus, no structure. Elongated, on edge.  There is a tiny bit missing in the segm
## 4
## 5
## 6
##   annotation_method  method
## 1      Segmentation microCT
## 2      Segmentation microCT
## 3      Segmentation microCT
## 4      Segmentation microCT
## 5      Segmentation microCT
## 6      Segmentation microCT
```

```r
summary(data)
```

```
##  specimen_id        species_replicate    phylum            species
##  Length:2073        Min.   :1.00      Length:2073        Length:2073
##  Class :character   1st Qu.:2.00      Class :character   Class :character
##  Mode  :character   Median :2.00      Mode  :character   Mode  :character
##                     Mean   :2.09
##                     3rd Qu.:2.00
##                     Max.   :3.00
##
##     stage               type            measurement      volume
##  Length:2073        Length:2073        Min.   :  1   Min.   :  0.0151
##  Class :character   Class :character   1st Qu.: 23   1st Qu.:  0.3668
##  Mode  :character   Mode  :character   Median : 85   Median :  0.5588
##                                        Mean   :172   Mean   : 28.2048
##                                        3rd Qu.:268   3rd Qu.:  0.9261
##                                        Max.   :708   Max.   :811.8048
##                                                      NA's   :1852
##  surface_area_lindblad_2005_mm2  aspect_ratio     feret_min_mm
##  Min.   : 0.0962                Min.   :0.3602   Min.   :0.0024
##  1st Qu.: 2.9506                1st Qu.:1.2317   1st Qu.:0.0055
##  Median : 4.4620                Median :1.4390   Median :0.0055
##  Mean   : 5.0455                Mean   :1.8867   Mean   :0.1489
##  3rd Qu.: 6.7422                3rd Qu.:2.1846   3rd Qu.:0.0055
##  Max.   :20.0356                Max.   :8.0039   Max.   :1.4896
##  NA's   :1865                   NA's   :1839     NA's   :798
##  feret_mean_mm     feret_max_mm     feret_micron        notes
##  Min.   :0.0432   Min.   :0.0549   Min.   :  21.75   Length:2073
##  1st Qu.:0.0936   1st Qu.:0.1235   1st Qu.: 104.62   Class :character
##  Median :0.1283   Median :0.1685   Median : 159.22   Mode  :character
##  Mean   :0.3053   Mean   :0.4057   Mean   : 379.71
##  3rd Qu.:0.2651   3rd Qu.:0.3941   3rd Qu.: 450.43
##  Max.   :1.8639   Max.   :2.5205   Max.   :2520.51
##  NA's   :798      NA's   :786      NA's   :17
##  annotation_method     method
##  Length:2073        Length:2073
##  Class :character   Class :character
##  Mode  :character   Mode  :character
##
##
##
##
```

```r
# Ensure Feret diameter is numeric
data$feret_micron <- as.numeric(data$feret_micron)

# SUBSET GAMETE DATA ONLY
# Keep only gamete-related measurements (exclude tissue classes like mesohyl/spicules)
gametes <- data %>%
  filter(grepl("Gamete", type, ignore.case = TRUE),
         !is.na(feret_micron))

# Add phylum column for broad taxonomic comparisons
```

```r
gametes <- gametes %>%
  mutate(
    phylum = ifelse(species == "Paragorgia arborea",
                    "Cnidaria",
                    "Porifera")
  )
```

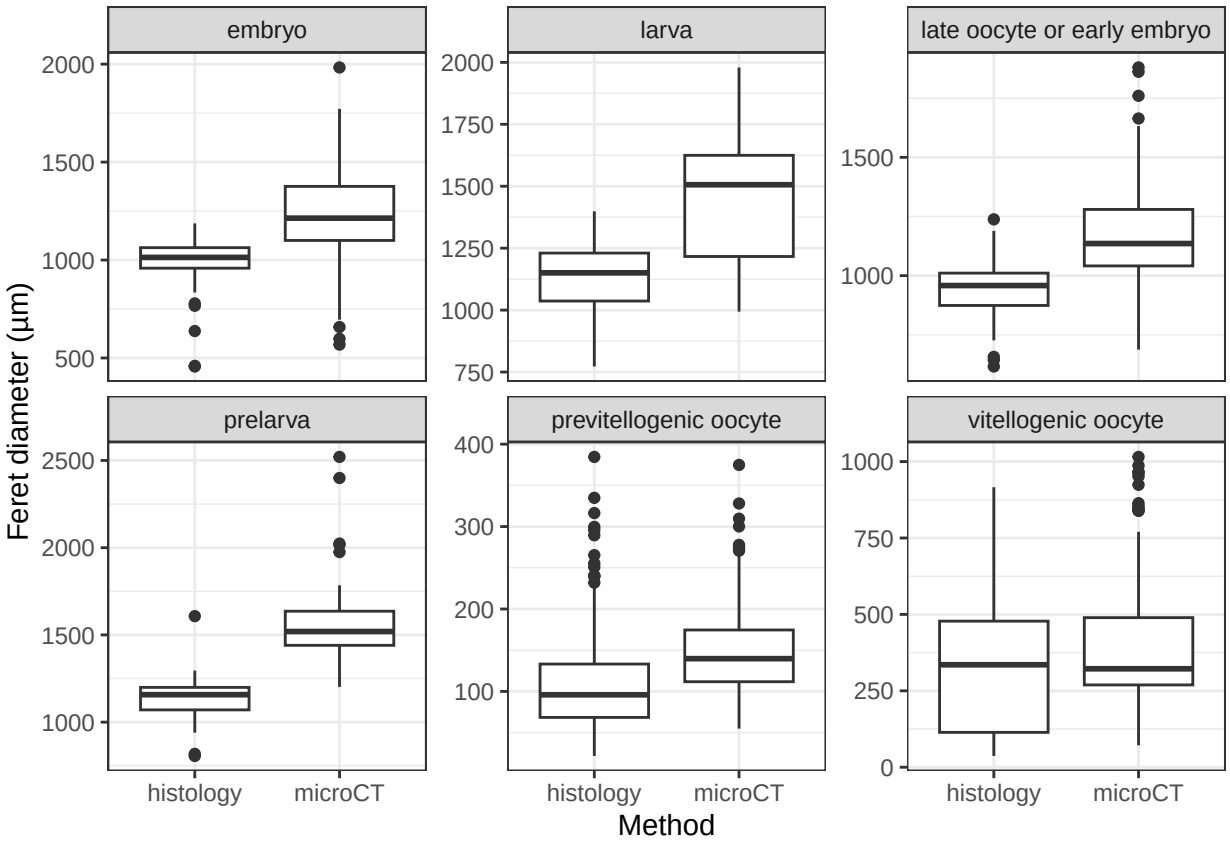
---

3. DESCRIPTIVE SUMMARY

```r
# Total gametes measured per method
gametes %>%
  group_by(method) %>%
  summarise(
    n = n(),
    mean_feret = mean(feret_micron, na.rm = TRUE),
    sd_feret = sd(feret_micron, na.rm = TRUE)
  )
```

```
## # A tibble: 2 x 4
##   method        n mean_feret sd_feret
##   <chr>     <int>      <dbl>    <dbl>
## 1 histology   769       336.     381.
## 2 microCT    1287       406.     468.
```

```r
# Raw Feret distributions by method and stage
ggplot(gametes, aes(x = method, y = feret_micron)) +
  geom_boxplot() +
  facet_wrap(~ stage, scales = "free_y") +
  theme_bw() +
  labs(x = "Method", y = "Feret diameter (µm)")
```



---

4. DETECTION LIMIT

```
# ---- 5a. MicroCT detection limits ----
# Overall minimum
microCT_limit <- gametes %>%
  filter(method == "microCT") %>%
  summarise(min_detected = min(feret_micron, na.rm = TRUE))
print(microCT_limit)
```

```
##   min_detected
## 1     54.90675
```

```
# By phylum
microCT_limit_sponges <- gametes %>%
  filter(method == "microCT", phylum == "Porifera") %>%
  summarise(min_detected = min(feret_micron, na.rm = TRUE))
print(microCT_limit_sponges)
```

```
##   min_detected
## 1     71.68071
```

```r
microCT_limit_cnidaria <- gametes %>%
  filter(method == "microCT", phylum == "Cnidaria") %>%
  summarise(min_detected = min(feret_micron, na.rm = TRUE))
print(microCT_limit_cnidaria)
```

```
##   min_detected
## 1     54.90675
```

```r
# ---- 5b. Histology detection limits ----
histo_limit <- gametes %>%
  filter(method == "histology") %>%
  summarise(min_detected = min(feret_micron, na.rm = TRUE))
print(histo_limit)
```

```
##   min_detected
## 1       21.746
```

```r
histo_limit_sponges <- gametes %>%
  filter(method == "histology", phylum == "Porifera") %>%
  summarise(min_detected = min(feret_micron, na.rm = TRUE))
print(histo_limit_sponges)
```

```
##   min_detected
## 1       21.746
```

```r
histo_limit_cnidaria <- gametes %>%
  filter(method == "histology", phylum == "Cnidaria") %>%
  summarise(min_detected = min(feret_micron, na.rm = TRUE))
print(histo_limit_cnidaria)
```

```
##   min_detected
## 1       24.403
```

---

5. GAMETE COUNTS

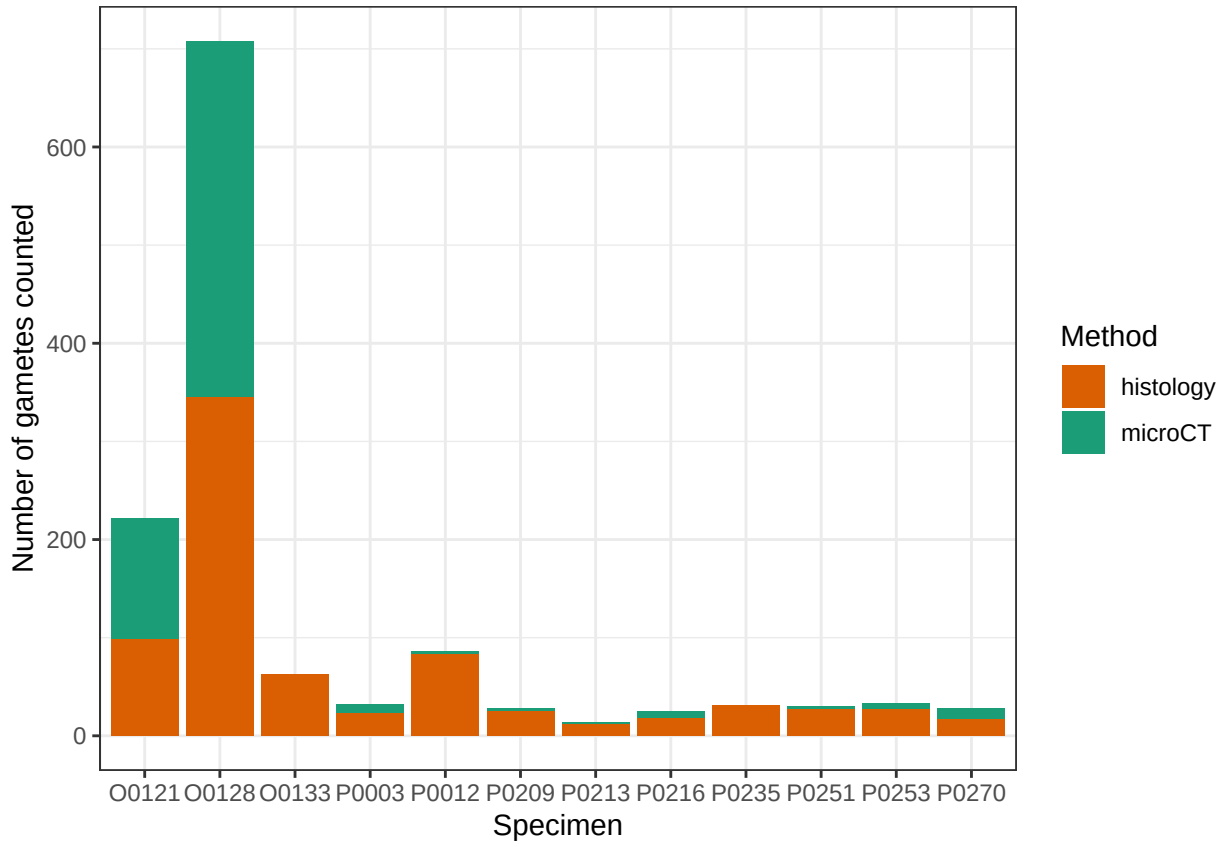
```r
# ---- 4a. Total counts per specimen and method ----
counts_table <- gametes %>%
  group_by(specimen_id, method) %>%
  summarise(n = n(), .groups = "drop") %>%
  pivot_wider(names_from = method, values_from = n, values_fill = 0)

print(counts_table)
```

```
## # A tibble: 12 x 3
##    specimen_id histology microCT
##    <chr>           <int>   <int>
```

```
##  1 O0121              98     222
##  2 O0128             345     708
##  3 O0133              63      60
##  4 P0003              23      32
##  5 P0012              83      86
##  6 P0209              25      28
##  7 P0213              12      14
##  8 P0216              18      25
##  9 P0235              31      21
## 10 P0251              27      30
## 11 P0253              27      33
## 12 P0270              17      28
```

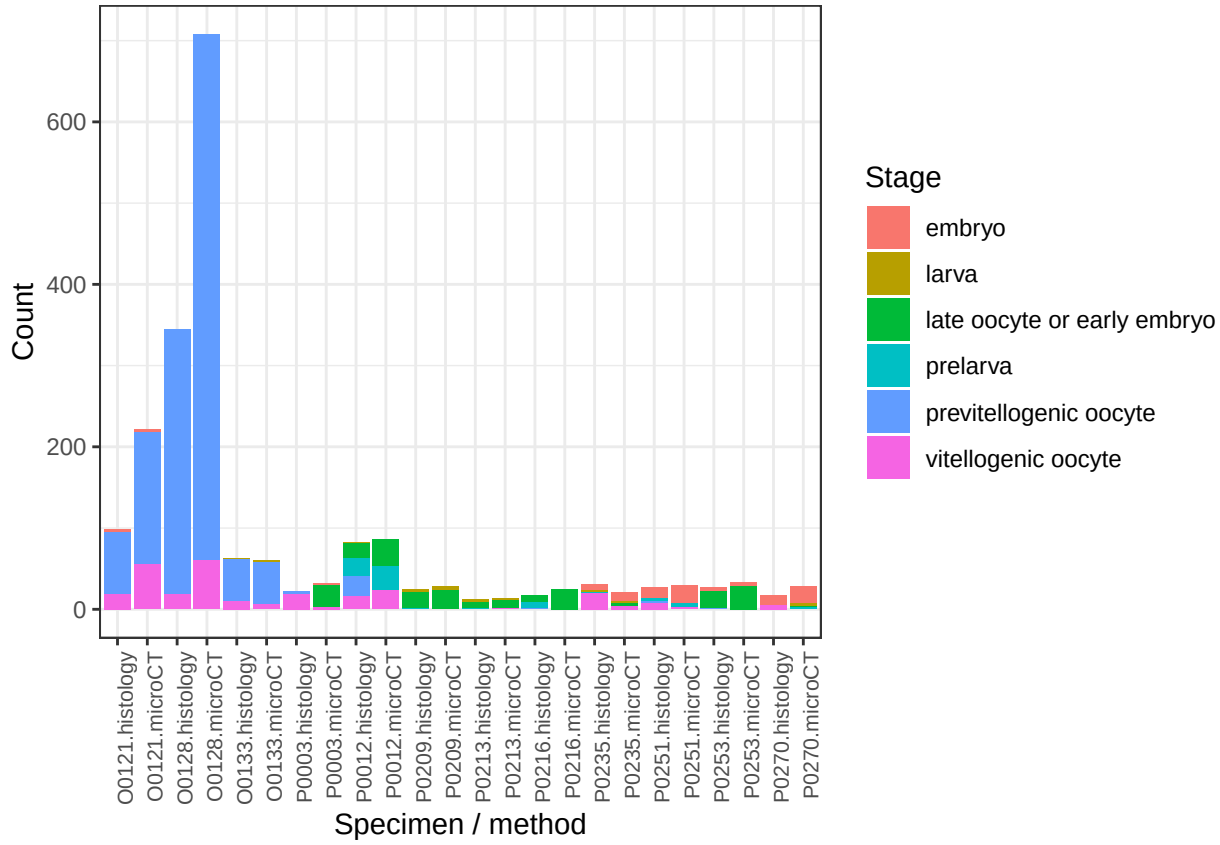
```r
# Bar plot: counts per specimen
ggplot(counts_table, aes(x = specimen_id)) +
  geom_col(aes(y = microCT, fill = "microCT"), position = position_dodge(0.9)) +
  geom_col(aes(y = histology, fill = "histology"), position = position_dodge(0.9)) +
  labs(y = "Number of gametes counted", x = "Specimen", fill = "Method") +
  theme_bw() +
  scale_fill_manual(values = c("microCT" = "#1b9e77", "histology" = "#d95f02"))
```




```r
# ---- 4b. Counts by stage (stacked) ----
counts_stage <- gametes %>%
  group_by(specimen_id, method, stage) %>%
  summarise(n = n(), .groups = "drop")
```

```r
# Create a combined label for x-axis
counts_stage <- counts_stage %>%
  mutate(specimen_method = paste(specimen_id, method, sep = "."))

ggplot(counts_stage, aes(x = specimen_method, y = n, fill = stage)) +
  geom_col() +
  theme_bw() +
  theme(
    axis.text.x = element_text(angle = 90, hjust = 1, size = 8)
  ) +
  labs(
    x = "Specimen / method",
    y = "Count",
    fill = "Stage"
  )
```




```r
# No stats between bars because I then only compare two numbers... I could group the species and then c

# ---- 4c. Ratio of microCT to histology counts ----
counts_table <- counts_table %>%
  mutate(ratio = ifelse(histology > 0, microCT / histology, NA))

counts_table %>%
  dplyr::select(specimen_id, microCT, histology, ratio) %>%
  print()
```

```
## # A tibble: 12 x 4
##    specimen_id microCT histology ratio
##    <chr>         <int>     <int> <dbl>
##  1 O0121           222        98 2.27
##  2 O0128           708       345 2.05
##  3 O0133            60        63 0.952
##  4 P0003            32        23 1.39
##  5 P0012            86        83 1.04
##  6 P0209            28        25 1.12
##  7 P0213            14        12 1.17
##  8 P0216            25        18 1.39
##  9 P0235            21        31 0.677
## 10 P0251            30        27 1.11
## 11 P0253            33        27 1.22
## 12 P0270            28        17 1.65
```

```
# Test if normal distribution or not: normality test (if p is below 0.05 then not non-parametric)
shapiro.test(counts_table$microCT) # = not normal
```

```
##
##  Shapiro-Wilk normality test
##
## data:  counts_table$microCT
## W = 0.50725, p-value = 1.999e-05
```

```
shapiro.test(counts_table$histology) # = not normal
```

```
##
##  Shapiro-Wilk normality test
##
## data:  counts_table$histology
## W = 0.56635, p-value = 5.7e-05
```

```
# ---- 4d. Paired statistical tests on counts ----
# Wilcoxon signed-rank test (non-parametric, robust to outliers)
wilcox.test(counts_table$microCT, counts_table$histology, paired = TRUE)
```

```
##
##  Wilcoxon signed rank test with continuity correction
##
## data:  counts_table$microCT and counts_table$histology
## V = 65.5, p-value = 0.04061
## alternative hypothesis: true location shift is not equal to 0
```

```
# ---- 4e. Phylum-specific count comparison ----
# Join phylum info to counts_table
counts_table <- counts_table %>%
  left_join(
    gametes %>% dplyr::select(specimen_id, phylum) %>% distinct(),
    by = "specimen_id"
  )
```

```r
# Split into phyla and test
counts_porifera <- counts_table %>% filter(phylum == "Porifera")
counts_cnidaria <- counts_table %>% filter(phylum == "Cnidaria")

# Wilcoxon tests for each phylum
print("Porifera counts (microCT vs histology):")
```

```
## [1] "Porifera counts (microCT vs histology):"
```

```r
wilcox.test(counts_porifera$microCT, counts_porifera$histology, paired = TRUE)
```

```
##
##  Wilcoxon signed rank test with continuity correction
##
## data:  counts_porifera$microCT and counts_porifera$histology
## V = 37, p-value = 0.09603
## alternative hypothesis: true location shift is not equal to 0
```

```r
print("Cnidaria counts (microCT vs histology):")
```

```
## [1] "Cnidaria counts (microCT vs histology):"
```

```r
wilcox.test(counts_cnidaria$microCT, counts_cnidaria$histology, paired = TRUE) #only three samples/numb
```

```
##
##  Wilcoxon signed rank exact test
##
## data:  counts_cnidaria$microCT and counts_cnidaria$histology
## V = 5, p-value = 0.5
## alternative hypothesis: true location shift is not equal to 0
```

6. COMPARE COUNTS (>70 µm DETECTION THRESHOLD)

```r
# ---- 6a. Filter ----
gametes_large <- gametes %>% filter(feret_micron > 70)

counts_large <- gametes_large %>%
  group_by(specimen_id, method) %>%
  summarise(n = n(), .groups = "drop") %>%
  pivot_wider(names_from = method, values_from = n, values_fill = 0) %>%
  mutate(
    ratio = ifelse(histology > 0, microCT / histology, NA)
  ) %>%
  # Add phylum for grouping
  left_join(
    gametes %>% dplyr::select(specimen_id, phylum, species) %>% distinct(),
    by = "specimen_id"
  )

print(counts_large)
```

```
## # A tibble: 12 x 6
##    specimen_id histology microCT ratio phylum   species
##    <chr>           <int>   <int> <dbl> <chr>    <chr>
##  1 O0121              76     222  2.92 Cnidaria Paragorgia arborea
##  2 O0128             283     703  2.48 Cnidaria Paragorgia arborea
##  3 O0133              51      59  1.16 Cnidaria Paragorgia arborea
##  4 P0003              17      32  1.88 Porifera Tedania kagalaskai
##  5 P0012              49      86  1.76 Porifera Tedania kagalaskai
##  6 P0209              25      28  1.12 Porifera Monanchora pulchra
##  7 P0213              12      14  1.17 Porifera Monanchora pulchra
##  8 P0216              17      25  1.47 Porifera Monanchora pulchra
##  9 P0235              25      21  0.84 Porifera Stelodoryx oxeata
## 10 P0251              25      30  1.2  Porifera Stelodoryx oxeata
## 11 P0253              26      33  1.27 Porifera Tedania kagalaskai
## 12 P0270              17      28  1.65 Porifera Stelodoryx oxeata
```

```
# ---- 6b. Statistical tests ----
# Overall (all taxa)
p_global <- wilcox.test(counts_large$microCT, counts_large$histology, paired = TRUE)$p.value
p_label_global <- ifelse(p_global < 0.001, "p < 0.001", paste0("p = ", round(p_global, 3)))

# Sponges only
counts_large_sponges <- counts_large %>% filter(phylum == "Porifera")
p_sponges <- t.test(counts_large_sponges$microCT, counts_large_sponges$histology, paired = TRUE)$p.valu
p_label_sponges <- ifelse(p_sponges < 0.001, "p < 0.001", paste0("p = ", round(p_sponges, 3)))

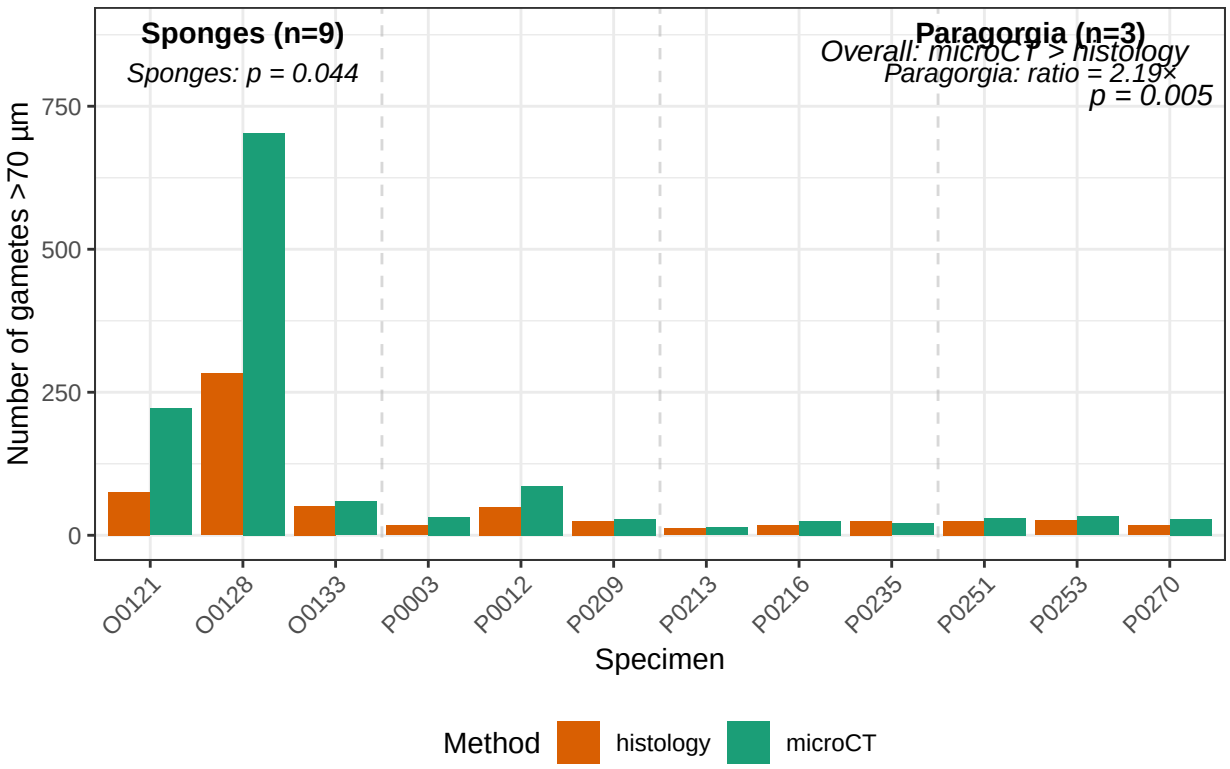
# Paragorgia only (ratio, no p-value due to n=3)
counts_large_coral <- counts_large %>% filter(phylum == "Cnidaria")
mean_ratio_coral <- mean(counts_large_coral$ratio, na.rm = TRUE)

# ---- 6c. Publication-ready bar plot with annotations ----
# Reshape for plotting
counts_large_plot <- counts_large %>%
  pivot_longer(cols = c(histology, microCT),
               names_to = "method",
               values_to = "n")

# Determine max value for annotation placement
max_n <- max(counts_large_plot$n, na.rm = TRUE)

ggplot(counts_large_plot, aes(x = specimen_id, y = n, fill = method)) +
  geom_col(position = position_dodge(0.9)) +
  # Add overall significance annotation
  annotate("text", x = Inf, y = Inf,
           label = paste0("Overall: microCT > histology\n", p_label_global),
           hjust = 1.1, vjust = 1.5, size = 4, fontface = "italic") +
  # Add sponges p-value
  annotate("text", x = 2, y = max_n * 1.15,
           label = paste0("Sponges: p = ", round(p_sponges, 3)),
           size = 3.5, fontface = "italic") +
  # Add Paragorgia ratio
  annotate("text", x = 10.5, y = max_n * 1.15,
           label = paste0("Paragorgia: ratio = ", round(mean_ratio_coral, 2), "×"),
           size = 3.5, fontface = "italic") +
```

```
  # Add dashed lines to separate phyla (optional)
  geom_vline(xintercept = c(3.5, 6.5, 9.5), linetype = "dashed", color = "grey70", alpha = 0.5) +
  # Add phylum labels
  annotate("text", x = 2, y = max_n * 1.25,
           label = "Sponges (n=9)", fontface = "bold", size = 4) +
  annotate("text", x = 10.5, y = max_n * 1.25,
           label = "Paragorgia (n=3)", fontface = "bold", size = 4) +
  scale_fill_manual(values = c("histology" = "#d95f02", "microCT" = "#1b9e77")) +
  theme_bw() +
  labs(
    x = "Specimen",
    y = "Number of gametes >70 µm",
    fill = "Method",
    caption = "Dashed lines separate phyla. Overall comparison: Wilcoxon signed-rank test.\nSponges: pa
  ) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "bottom"
  )
```

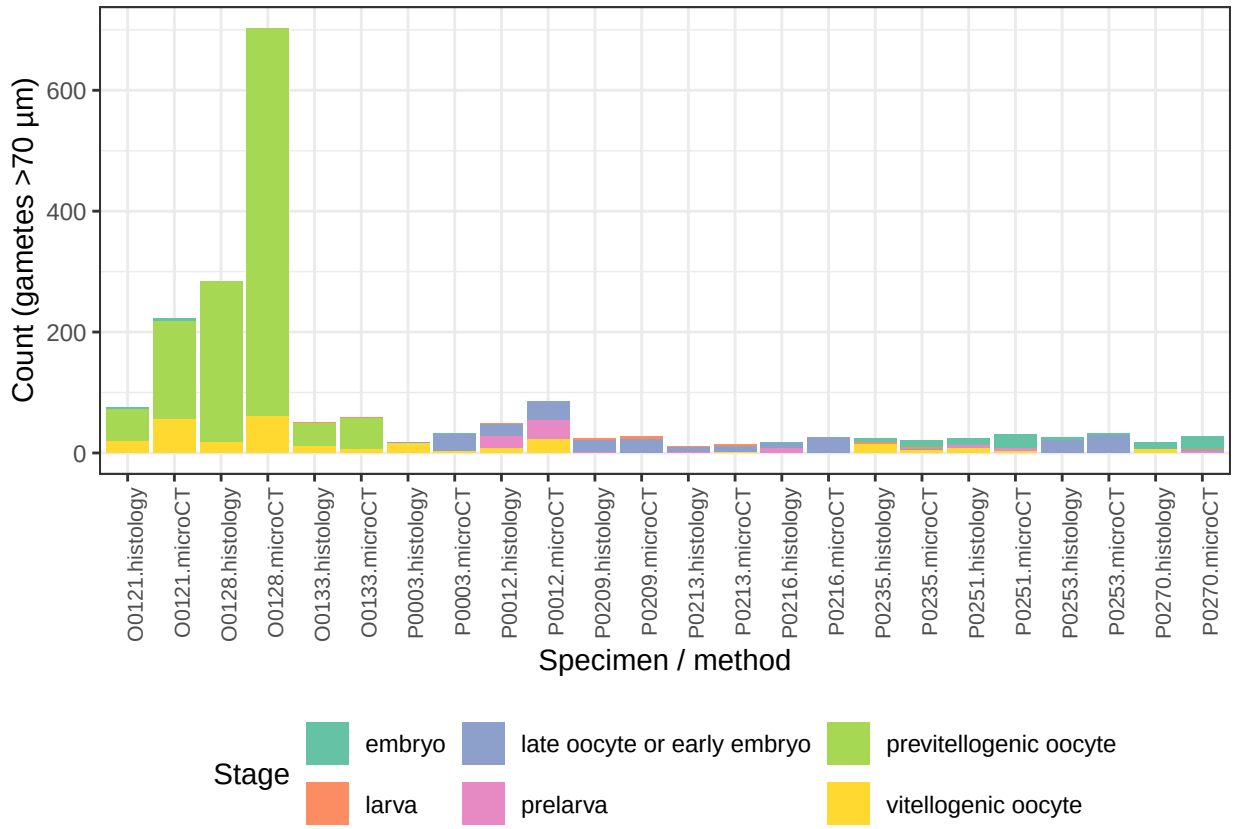


Dashed lines separate phyla. Overall comparison: Wilcoxon signed−rank test.
Sponges: paired t−test. Paragorgia: n=3, ratio shown instead of p−value.

```
# ---- Filtered counts by stage (stacked) ----
# Use gametes_large (already filtered to >70 µm)
counts_stage_filtered <- gametes_large %>%
  group_by(specimen_id, method, stage) %>%
  summarise(n = n(), .groups = "drop")
```

```r
# Create a combined label for x-axis
counts_stage_filtered <- counts_stage_filtered %>%
  mutate(specimen_method = paste(specimen_id, method, sep = "."))

# Stacked bar plot
ggplot(counts_stage_filtered, aes(x = specimen_method, y = n, fill = stage)) +
  geom_col() +
  theme_bw() +
  theme(
    axis.text.x = element_text(angle = 90, hjust = 1, size = 8),
    legend.position = "bottom"
  ) +
  labs(
    x = "Specimen / method",
    y = "Count (gametes >70 µm)",
    fill = "Stage"
  ) +
  scale_fill_brewer(palette = "Set2", name = "Stage")
```




```r
# ---- 6d. Sponge-only tests on filtered counts ----
# Subset to Porifera only
counts_large_porifera <- counts_large %>% filter(phylum == "Porifera")

shapiro.test(counts_large_porifera$microCT) #not normal so use Wilcox
```

```
##
##  Shapiro-Wilk normality test
##
## data:  counts_large_porifera$microCT
## W = 0.66605, p-value = 0.0005689
```

```
shapiro.test(counts_large_porifera$histology) #not normal so use Wilcox
```

```
##
##  Shapiro-Wilk normality test
##
## data:  counts_large_porifera$histology
## W = 0.79333, p-value = 0.01713
```

```
#Wilcox paired test (non parametric)
wilcox_test_sponges <- wilcox.test(counts_large_porifera$microCT,
                                   counts_large_porifera$histology,
                                   paired = TRUE)
print(wilcox_test_sponges)
```

```
##
##  Wilcoxon signed rank exact test
##
## data:  counts_large_porifera$microCT and counts_large_porifera$histology
## V = 42, p-value = 0.01953
## alternative hypothesis: true location shift is not equal to 0
```

```
# Paired t-test (parametric, robust with n=9) - just to check if it fits with Wilcox test,
t_test_sponges <- t.test(counts_large_porifera$microCT,
                         counts_large_porifera$histology,
                         paired = TRUE)
print(t_test_sponges)
```

```
##
##  Paired t-test
##
## data:  counts_large_porifera$microCT and counts_large_porifera$histology
## t = 2.39, df = 8, p-value = 0.04385
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##   0.3280993 18.3385674
## sample estimates:
## mean difference
##        9.333333
```

```
# ---- View ratios from filtered data ----
counts_large %>%
  dplyr::select(specimen_id, phylum, species, histology, microCT, ratio) %>%
  arrange(phylum, desc(ratio)) %>%
  print()
```

```
## # A tibble: 12 x 6
##    specimen_id phylum   species            histology microCT ratio
##    <chr>       <chr>    <chr>                  <int>   <int> <dbl>
##  1 O0121       Cnidaria Paragorgia arborea        76     222  2.92
##  2 O0128       Cnidaria Paragorgia arborea       283     703  2.48
##  3 O0133       Cnidaria Paragorgia arborea        51      59  1.16
##  4 P0003       Porifera Tedania kagalaskai        17      32  1.88
##  5 P0012       Porifera Tedania kagalaskai        49      86  1.76
##  6 P0270       Porifera Stelodoryx oxeata         17      28  1.65
##  7 P0216       Porifera Monanchora pulchra        17      25  1.47
##  8 P0253       Porifera Tedania kagalaskai        26      33  1.27
##  9 P0251       Porifera Stelodoryx oxeata         25      30  1.2
## 10 P0213       Porifera Monanchora pulchra        12      14  1.17
## 11 P0209       Porifera Monanchora pulchra        25      28  1.12
## 12 P0235       Porifera Stelodoryx oxeata         25      21  0.84
```

---

7. PAIRED COMPARISON: SPECIMEN × STAGE × METHOD MEANS

```r
# Aggregate to specimen × stage × method means
# This is the unit of analysis for paired comparisons
specimen_stage <- gametes %>%
  group_by(specimen_id, species, phylum, stage, method) %>%
  summarise(
    n = n(),
    mean_feret = mean(feret_micron, na.rm = TRUE),
    median_feret = median(feret_micron, na.rm = TRUE),
    sd_feret = sd(feret_micron, na.rm = TRUE),
    .groups = "drop"
  )
```

---

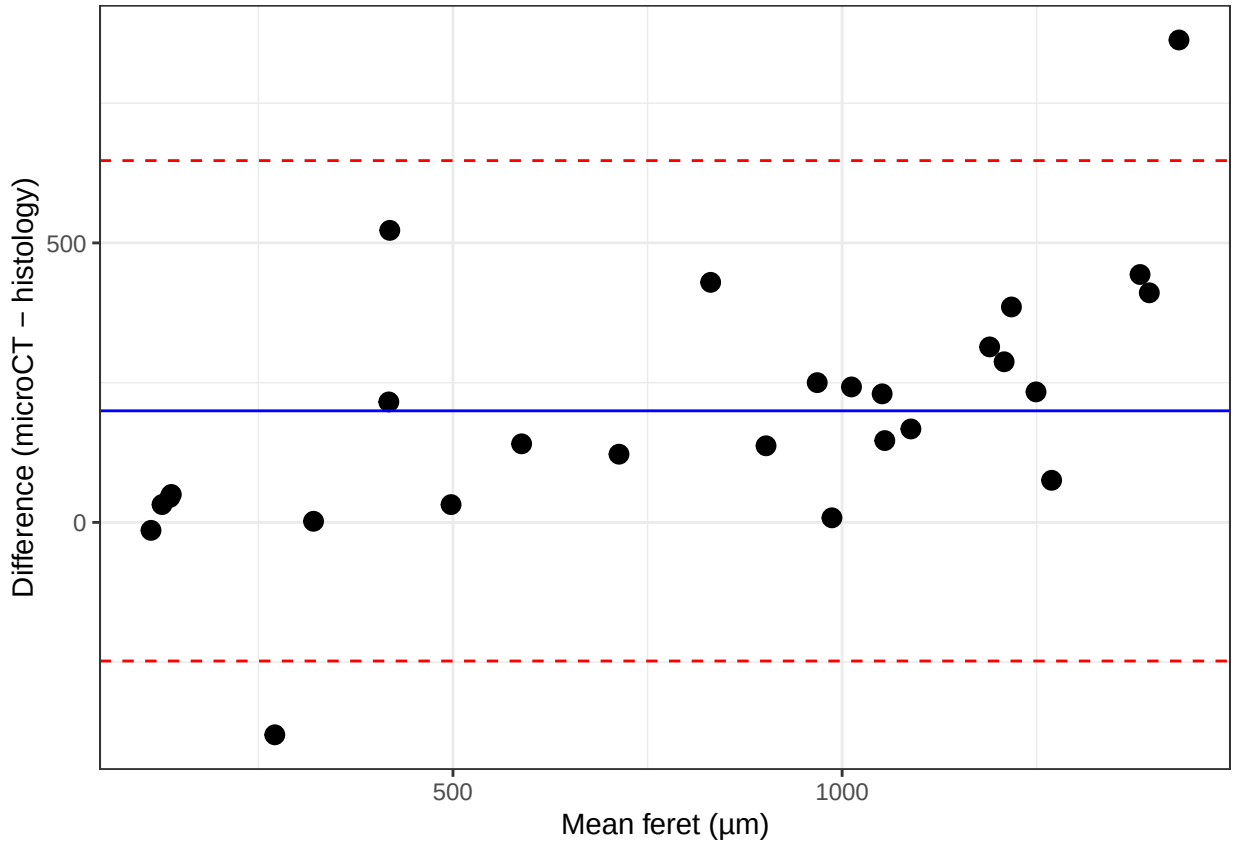
8. UNFILTERED ANALYSIS (MAIN MODEL)

```r
# ---- 8a. Bland-Altman (unfiltered) ----

paired_raw <- specimen_stage %>%
  dplyr::select(specimen_id, species, stage, method, mean_feret) %>%
  pivot_wider(names_from = method, values_from = mean_feret) %>%
  filter(!is.na(microCT) & !is.na(histology)) %>%
  mutate(diff = microCT - histology,
         mean = (microCT + histology) / 2)

# Bland-Altman plot - this is to see how the methods differ.
ggplot(paired_raw, aes(x = mean, y = diff)) +
  geom_point(size = 3) +
  geom_hline(yintercept = mean(paired_raw$diff), linetype = "solid", color = "blue") +
```

```
  geom_hline(yintercept = mean(paired_raw$diff) + 1.96 * sd(paired_raw$diff),
             linetype = "dashed", color = "red") +
  geom_hline(yintercept = mean(paired_raw$diff) - 1.96 * sd(paired_raw$diff),
             linetype = "dashed", color = "red") +
  labs(x = "Mean feret (µm)", y = "Difference (microCT - histology)") +
  theme_bw()
```

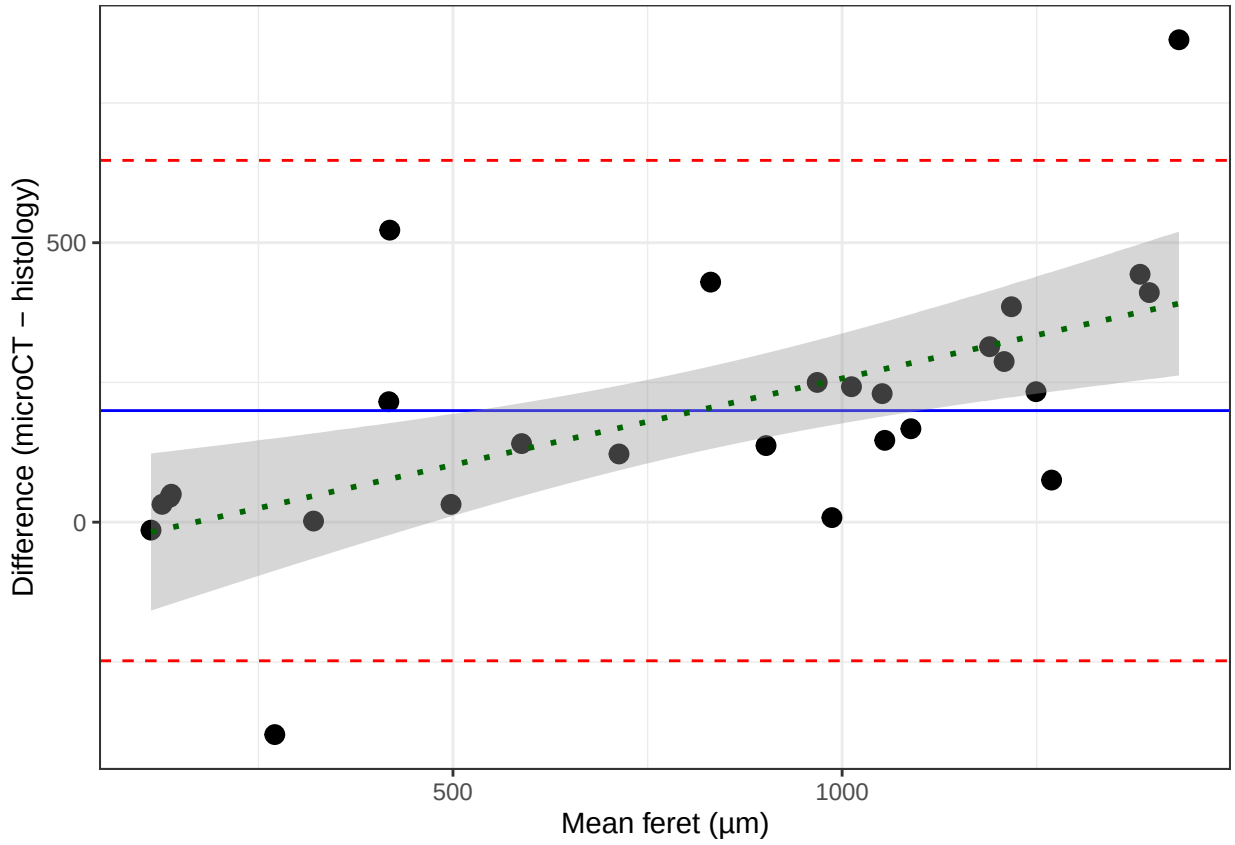



```
# Proportional bias test (unfiltered)
prop_bias <- lm(diff ~ mean, data = paired_raw) #fitting a linear regression (response = diff, microCT
summary(prop_bias) # This tests whether the difference between methods changes with the size of the mea
```

```
##
## Call:
## lm(formula = diff ~ mean, data = paired_raw)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -411.63  -79.86   -1.90   57.17  471.96
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -52.31885   76.27902  -0.686 0.499096
## mean          0.30951    0.08269   3.743 0.000955 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 186.5 on 25 degrees of freedom
## Multiple R-squared:  0.3592, Adjusted R-squared:  0.3335
## F-statistic: 14.01 on 1 and 25 DF,  p-value: 0.0009554
```

```r
# Visualise the proportional bias: Bland-Altman plot with proportional bias line
ggplot(paired_raw, aes(x = mean, y = diff)) +
  geom_point(size = 3) +
  geom_hline(yintercept = mean(paired_raw$diff), linetype = "solid", color = "blue") +
  geom_hline(yintercept = mean(paired_raw$diff) + 1.96 * sd(paired_raw$diff),
             linetype = "dashed", color = "red") +
  geom_hline(yintercept = mean(paired_raw$diff) - 1.96 * sd(paired_raw$diff),
             linetype = "dashed", color = "red") +
  geom_smooth(method = "lm", se = TRUE, color = "darkgreen", linetype = "dotted") +  # <-- ADD THIS
  labs(x = "Mean feret (µm)", y = "Difference (microCT - histology)") +
  theme_bw()
```




```r
# ---- Bland-Altman summary statistics ----
mean_diff <- mean(paired_raw$diff, na.rm = TRUE)
sd_diff <- sd(paired_raw$diff, na.rm = TRUE)
loa_lower <- mean_diff - 1.96 * sd_diff
loa_upper <- mean_diff + 1.96 * sd_diff

cat("Mean difference (microCT - histology):", round(mean_diff, 1), "µm\n")
```

```
## Mean difference (microCT - histology): 199.6 µm
```

```r
cat("SD of differences:", round(sd_diff, 1), "µm\n")
```

```
## SD of differences: 228.4 µm
```

```r
cat("95% limits of agreement: [", round(loa_lower, 1), ",", round(loa_upper, 1), "] µm\n")
```

```
## 95% limits of agreement: [ -248 , 647.3 ] µm
```

```r
# ---- 8b. Mixed-effects model (unfiltered) ---- MAIN MODEL

# Main model = liear mixed effects model (method × stage interaction with random intercept per specimen
# This tests: Main effect of method: Are microCT measurements systematically different from histology? 

model4 <- lmer(feret_micron ~ method * stage + (1 | specimen_id), data = gametes) #The random intercept
summary(model4)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: feret_micron ~ method * stage + (1 | specimen_id)
##    Data: gametes
##
## REML criterion at convergence: 25685.2
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -6.0901 -0.3858 -0.0615  0.3060  7.4358
##
## Random effects:
##  Groups      Name        Variance Std.Dev.
##  specimen_id (Intercept) 11275    106.2
##  Residual                16026    126.6
## Number of obs: 2056, groups:  specimen_id, 12
##
## Fixed effects:
##                                              Estimate Std. Error      df
## (Intercept)                                    959.07      37.44   21.54
## methodmicroCT                                  252.26      25.20 2033.19
## stagelarva                                     218.94      42.55 2043.92
## stagelate oocyte or early embryo                37.93      28.25 2009.87
## stageprelarva                                  178.76      31.45 2043.09
## stageprevitellogenic oocyte                   -957.53      25.45 2001.75
## stagevitellogenic oocyte                      -682.62      24.91 2042.68
## methodmicroCT:stagelarva                        72.88      53.70 2033.32
## methodmicroCT:stagelate oocyte or early embryo  -45.57      30.98 2034.50
## methodmicroCT:stageprelarva                    167.48      39.17 2034.36
## methodmicroCT:stageprevitellogenic oocyte      -216.58      26.23 2033.12
## methodmicroCT:stagevitellogenic oocyte         -241.64      30.36 2038.65
##                                              t value Pr(>|t|)
## (Intercept)                                   25.615  < 2e-16 ***
## methodmicroCT                                 10.009  < 2e-16 ***
## stagelarva                                     5.145 2.93e-07 ***
```

```
## stagelate oocyte or early embryo                 1.343    0.179
## stageprelarva                                    5.684 1.50e-08 ***
## stageprevitellogenic oocyte                    -37.626  < 2e-16 ***
## stagevitellogenic oocyte                       -27.399  < 2e-16 ***
## methodmicroCT:stagelarva                         1.357    0.175
## methodmicroCT:stagelate oocyte or early embryo  -1.471    0.142
## methodmicroCT:stageprelarva                      4.276 1.99e-05 ***
## methodmicroCT:stageprevitellogenic oocyte       -8.258 2.63e-16 ***
## methodmicroCT:stagevitellogenic oocyte          -7.959 2.85e-15 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##                 (Intr) mthdCT stglrv stooee stgprl stgpro stgvto mthdmcrCT:stgl
## methodmcrCT     -0.395
## stagelarva      -0.310  0.345
## stgltocyoee     -0.468  0.511  0.463
## stageprelrv     -0.404  0.467  0.378  0.594
## stgprvtllgo     -0.510  0.582  0.460  0.668  0.613
## stgvtllgnco     -0.490  0.599  0.446  0.651  0.595  0.816
## mthdmcrCT:stgl   0.202 -0.466 -0.722 -0.273 -0.249 -0.307 -0.307
## mthdCT:ooee      0.326 -0.809 -0.299 -0.684 -0.385 -0.472 -0.476  0.379
## mthdmcrCT:stgp   0.265 -0.639 -0.234 -0.354 -0.676 -0.373 -0.387  0.299
## mthdmcrCT:stgpo  0.380 -0.961 -0.333 -0.496 -0.454 -0.598 -0.572  0.448
## mthdmcrCT:stgvo  0.307 -0.834 -0.268 -0.396 -0.359 -0.387 -0.643  0.381
##                 mCooee mthdmcrCT:stgp mthdmcrCT:stgpo
## methodmcrCT
## stagelarva
## stgltocyoee
## stageprelrv
## stgprvtllgo
## stgvtllgnco
## mthdmcrCT:stgl
## mthdCT:ooee
## mthdmcrCT:stgp   0.515
## mthdmcrCT:stgpo  0.778  0.614
## mthdmcrCT:stgvo  0.666  0.535         0.806
```

```
anova(model4, type = 3) #type 3 because it tests each term after all otheer terms, including the intera
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                  Sum Sq  Mean Sq NumDF  DenDF  F value    Pr(>F)
## method          5599192  5599192     1 2037.4  349.375 < 2.2e-16 ***
## stage         102181796 20436359     5 2000.5 1275.174 < 2.2e-16 ***
## method:stage    4777497   955499     5 2037.0   59.621 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

STAGE SPECIFIC CONTRASTS (UNFILTERED DATA)

```r
# ---- 8c. Estimated marginal means and pairwise contrasts for stages ----
# This is a post-hoc pairwise comparison - the stage-specific contrasts that form the core of the stage

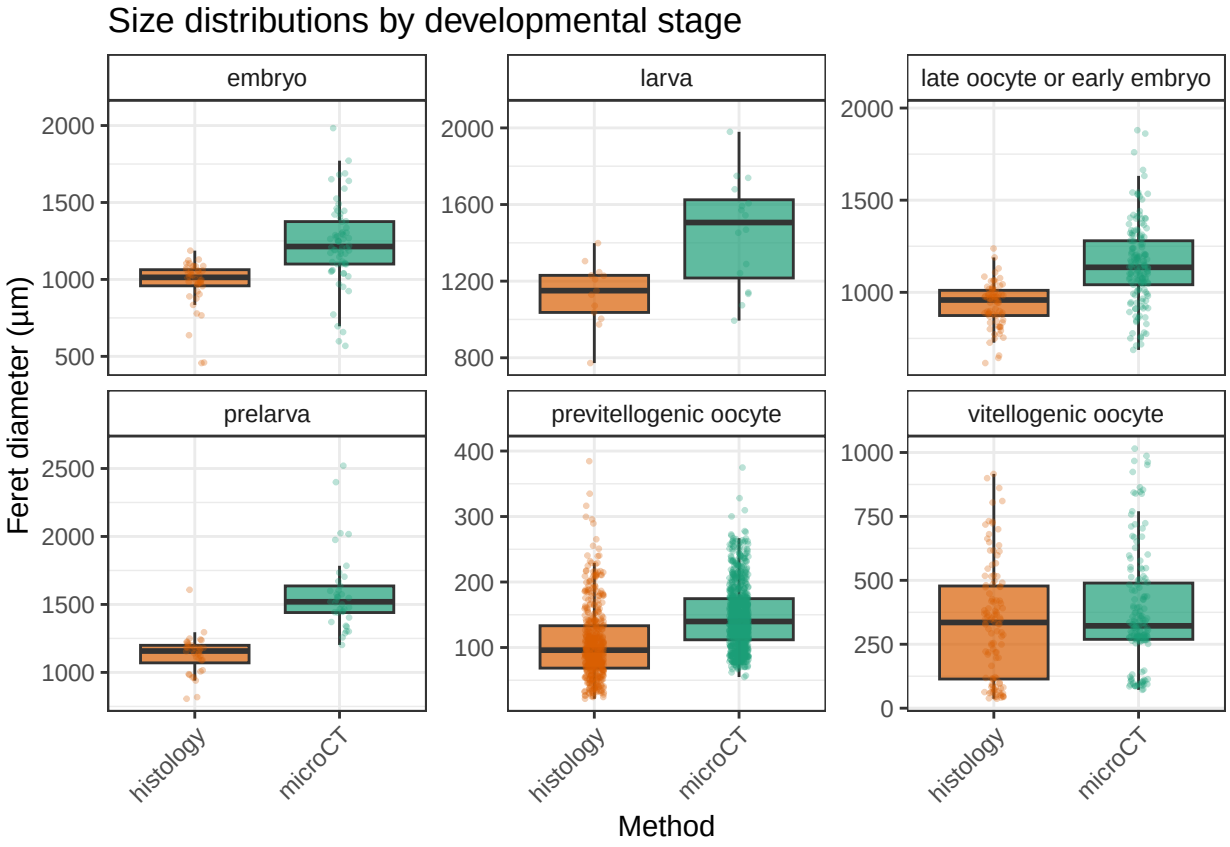
emm_raw <- emmeans(model4, ~ method | stage)
contrasts_raw <- contrast(emm_raw, "pairwise", by = "stage", adjust = "tukey")
print(contrasts_raw)
```

```
## stage = embryo:
##  contrast            estimate    SE   df t.ratio p.value
##  histology - microCT   -252.3 25.20 2034 -10.009 <0.0001
##
## stage = larva:
##  contrast            estimate    SE   df t.ratio p.value
##  histology - microCT   -325.1 47.50 2035  -6.842 <0.0001
##
## stage = late oocyte or early embryo:
##  contrast            estimate    SE   df t.ratio p.value
##  histology - microCT   -206.7 18.20 2040 -11.354 <0.0001
##
## stage = prelarva:
##  contrast            estimate    SE   df t.ratio p.value
##  histology - microCT   -419.7 30.10 2037 -13.925 <0.0001
##
## stage = previtellogenic oocyte:
##  contrast            estimate    SE   df t.ratio p.value
##  histology - microCT    -35.7  7.27 2035  -4.911 <0.0001
##
## stage = vitellogenic oocyte:
##  contrast            estimate    SE   df t.ratio p.value
##  histology - microCT    -10.6 16.80 2044  -0.632  0.5272
##
## Degrees-of-freedom method: kenward-roger
```

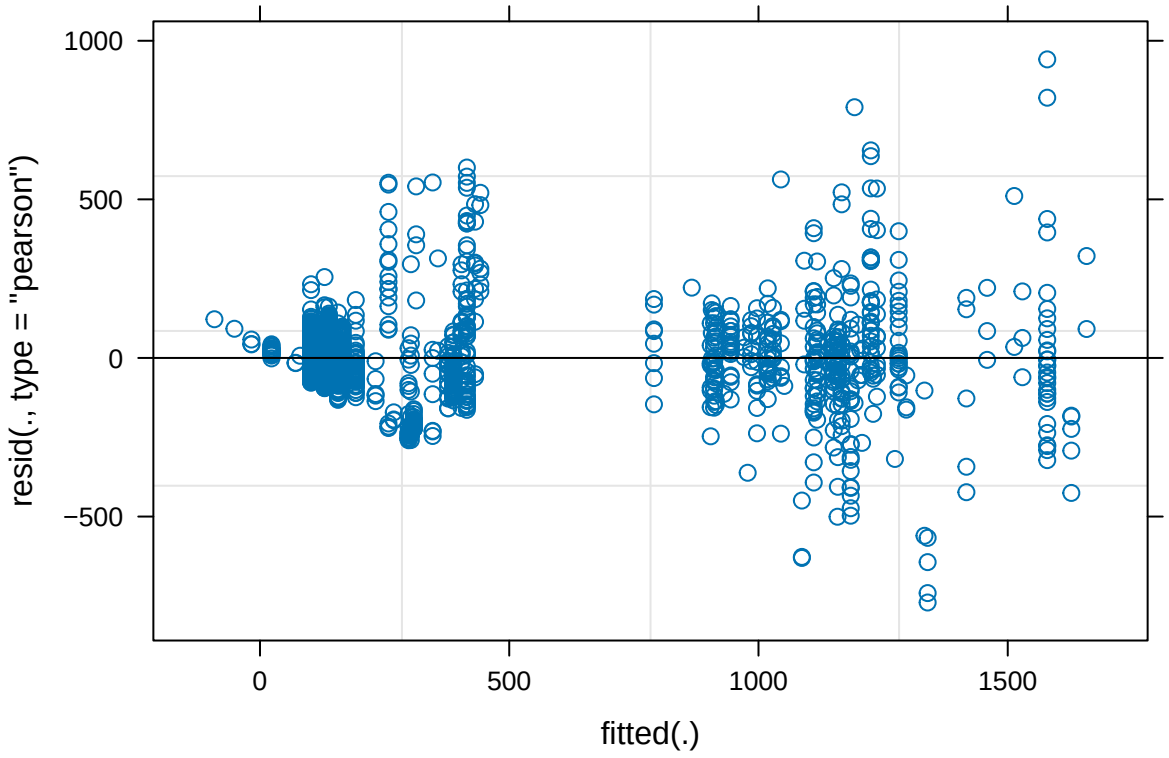
```r
# ---- Improved boxplot by stage with significance stars ----
# Extract significance from contrasts_raw
sig_data <- as.data.frame(contrasts_raw)
sig_data <- sig_data %>%
  mutate(
    star = case_when(
      p.value < 0.001 ~ "***",
      p.value < 0.01  ~ "**",
      p.value < 0.05  ~ "*",
      TRUE ~ "ns"
    ),
    # Calculate y position for stars (max Feret per stage + some offset)
    y_pos = gametes %>%
      group_by(stage) %>%
      summarise(max_feret = max(feret_micron, na.rm = TRUE)) %>%
      pull(max_feret) * 1.05
  )

# Plot
ggplot(gametes, aes(x = method, y = feret_micron, fill = method)) +
```

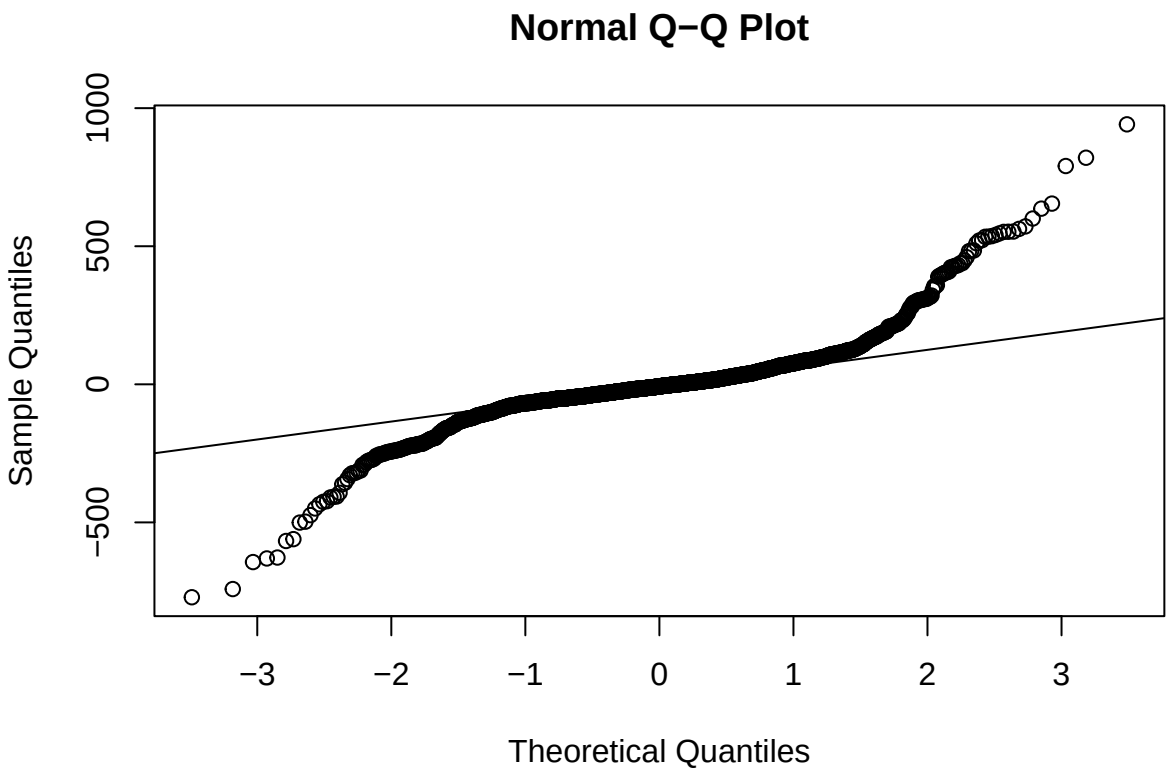
```r
  geom_boxplot(outlier.shape = NA, alpha = 0.7) +  # hide outliers for cleaner look
  geom_point(position = position_jitterdodge(jitter.width = 0.2),
             aes(color = method), size = 0.5, alpha = 0.3) +
  facet_wrap(~ stage, scales = "free_y") +
  # Add significance stars
  geom_text(data = sig_data %>% filter(star != "ns"),
            aes(x = 1.5, y = y_pos, label = star),
            inherit.aes = FALSE, size = 5, vjust = 0) +
  scale_fill_manual(values = c("histology" = "#d95f02", "microCT" = "#1b9e77")) +
  scale_color_manual(values = c("histology" = "#d95f02", "microCT" = "#1b9e77")) +
  theme_bw() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    strip.background = element_rect(fill = "white"),
    legend.position = "none"
  ) +
  labs(x = "Method", y = "Feret diameter (µm)",
       title = "Size distributions by developmental stage")
```




```r
# ---- 8d. Model diagnostics ----
plot(model4)          # residuals vs fitted
```




```
qqnorm(resid(model4)); qqline(resid(model4))
```



9. FILTERED ANALYSIS: FILTERED TO >70 µm

```
### Do the same as above, but with the filtered data

# ---- Combined stages with filtered data ----

# Step 1: Recode stages into combined categories
gametes_filtered <- gametes %>%
  filter(feret_micron > 70) %>%
  mutate(
    stage_combined = case_when(
      # Combine vitellogenic oocyte + late oocyte/early embryo (for sponges, since it was difficult to
      stage %in% c("vitellogenic oocyte", "late oocyte or early embryo") ~ "Vitellogenic / late oocyte"
      # Combine prelarva + larva (same, difficult to draw a line...)
      stage %in% c("prelarva", "larva") ~ "Larva / prelarva",
      # Keep others as is
      TRUE ~ stage
    )
  )

# Step 2: Refit mixed model with combined stages (= make stats matching)
# This is optional but recommended if the significance of stars should reflect the combined categories
model_combined <- lmer(feret_micron ~ method * stage_combined + (1 | specimen_id),
```

```
                      data = gametes_filtered)
summary(model_combined)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: feret_micron ~ method * stage_combined + (1 | specimen_id)
##    Data: gametes_filtered
##
## REML criterion at convergence: 25223
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.8494 -0.2452 -0.0191  0.2993  5.2563
##
## Random effects:
##  Groups      Name        Variance Std.Dev.
##  specimen_id (Intercept) 44019    209.8
##  Residual                33486    183.0
## Number of obs: 1904, groups:  specimen_id, 12
##
## Fixed effects:
##                                                    Estimate Std. Error
## (Intercept)                                          999.54      68.13
## methodmicroCT                                        274.61      36.42
## stage_combinedLarva / prelarva                        18.38      42.45
## stage_combinedprevitellogenic oocyte                -562.23      37.40
## stage_combinedVitellogenic / late oocyte            -384.49      35.01
## methodmicroCT:stage_combinedLarva / prelarva         175.38      51.58
## methodmicroCT:stage_combinedprevitellogenic oocyte  -251.26      38.21
## methodmicroCT:stage_combinedVitellogenic / late oocyte  -97.04      40.58
##                                                          df t value Pr(>|t|)
## (Intercept)                                           17.07  14.670 4.15e-11
## methodmicroCT                                       1885.35   7.539 7.30e-14
## stage_combinedLarva / prelarva                      1895.20   0.433 0.665148
## stage_combinedprevitellogenic oocyte                1893.57 -15.034  < 2e-16
## stage_combinedVitellogenic / late oocyte            1895.92 -10.982  < 2e-16
## methodmicroCT:stage_combinedLarva / prelarva        1885.64   3.400 0.000688
## methodmicroCT:stage_combinedprevitellogenic oocyte  1885.33  -6.575 6.28e-11
## methodmicroCT:stage_combinedVitellogenic / late oocyte 1886.22  -2.392 0.016873
##
## (Intercept)                                         ***
## methodmicroCT                                       ***
## stage_combinedLarva / prelarva
## stage_combinedprevitellogenic oocyte                ***
## stage_combinedVitellogenic / late oocyte            ***
## methodmicroCT:stage_combinedLarva / prelarva        ***
## methodmicroCT:stage_combinedprevitellogenic oocyte  ***
## methodmicroCT:stage_combinedVitellogenic / late oocyte *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) mthdCT st_L/p stg_co s_V/lo mCT:/p mCT:_o
```

```
## methodmcrCT -0.315
## stg_cmbnL/p -0.348  0.502
## stg_cmbndpo -0.411  0.568  0.663
## stg_cmbV/lo -0.418  0.609  0.699  0.829
## mthdCT:_L/p  0.234 -0.702 -0.690 -0.425 -0.453
## mthdmcCT:_o  0.300 -0.953 -0.479 -0.608 -0.581  0.669
## mthCT:_V/lo  0.271 -0.899 -0.429 -0.467 -0.658  0.627  0.858
```

```
anova(model_combined, type = 3)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                          Sum Sq  Mean Sq NumDF  DenDF F value    Pr(>F)
## method                  9186339  9186339     1 1886.0 274.332 < 2.2e-16 ***
## stage_combined         36315439 12105146     3 1891.4 361.496 < 2.2e-16 ***
## method:stage_combined   5938310  1979437     3 1886.6  59.112 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

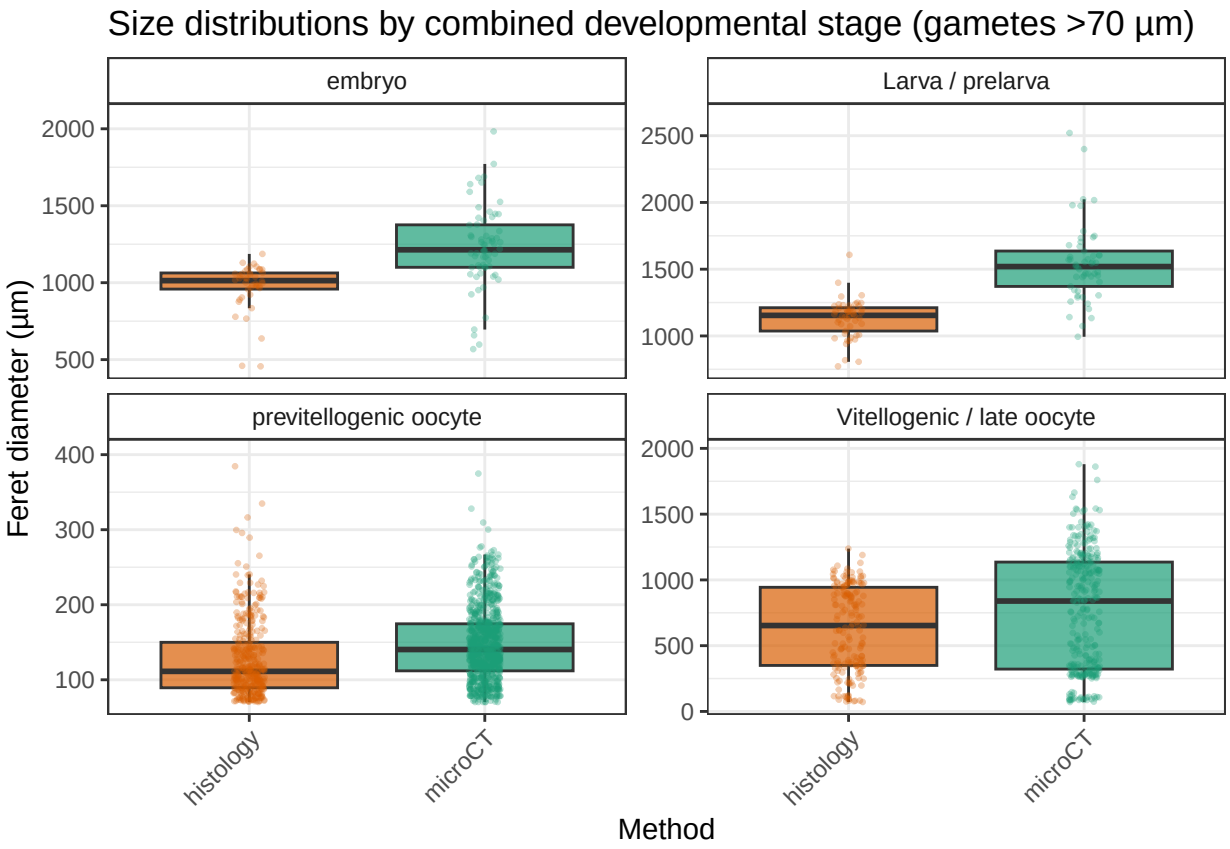
```
# Step 3: Get contrasts for combined stages
emm_combined <- emmeans(model_combined, ~ method | stage_combined)
contrasts_combined <- contrast(emm_combined, "pairwise", by = "stage_combined", adjust = "tukey")
print(contrasts_combined)
```

```
## stage_combined = embryo:
##  contrast            estimate   SE   df t.ratio p.value
##  histology - microCT   -274.6 36.4 1885  -7.539 <0.0001
##
## stage_combined = Larva / prelarva:
##  contrast            estimate   SE   df t.ratio p.value
##  histology - microCT   -450.0 36.8 1887 -12.243 <0.0001
##
## stage_combined = previtellogenic oocyte:
##  contrast            estimate   SE   df t.ratio p.value
##  histology - microCT    -23.3 11.6 1885  -2.020  0.0435
##
## stage_combined = Vitellogenic / late oocyte:
##  contrast            estimate   SE   df t.ratio p.value
##  histology - microCT   -177.6 17.7 1888 -10.007 <0.0001
##
## Degrees-of-freedom method: kenward-roger
```

```
# Step 4: Prepare significance stars for the plot
sig_data_combined <- as.data.frame(contrasts_combined)
sig_data_combined <- sig_data_combined %>%
  mutate(
    star = case_when(
      p.value < 0.001 ~ "***",
      p.value < 0.01  ~ "**",
      p.value < 0.05  ~ "*",
      TRUE ~ "ns"
    )
  ) %>%
```

```r
  # Get y-position for stars
  left_join(
    gametes_filtered %>%
      group_by(stage_combined) %>%
      summarise(max_feret = max(feret_micron, na.rm = TRUE), .groups = "drop"),
    by = "stage_combined"
  ) %>%
  mutate(y_pos = max_feret * 1.05)

# Step 5: Plot
ggplot(gametes_filtered, aes(x = method, y = feret_micron, fill = method)) +
  geom_boxplot(outlier.shape = NA, alpha = 0.7) +
  geom_point(position = position_jitterdodge(jitter.width = 0.2),
             aes(color = method), size = 0.5, alpha = 0.3) +
  facet_wrap(~ stage_combined, scales = "free_y") +
  # Add significance stars
  geom_text(data = sig_data_combined %>% filter(star != "ns"),
            aes(x = 1.5, y = y_pos, label = star),
            inherit.aes = FALSE, size = 5, vjust = 0) +
  # Add "ns" for non-significant stages
  geom_text(data = sig_data_combined %>% filter(star == "ns"),
            aes(x = 1.5, y = y_pos, label = "ns"),
            inherit.aes = FALSE, size = 4, vjust = 0, color = "grey40") +
  scale_fill_manual(values = c("histology" = "#d95f02", "microCT" = "#1b9e77")) +
  scale_color_manual(values = c("histology" = "#d95f02", "microCT" = "#1b9e77")) +
  theme_bw() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    strip.background = element_rect(fill = "white"),
    legend.position = "none"
  ) +
  labs(x = "Method", y = "Feret diameter (µm)",
       title = "Size distributions by combined developmental stage (gametes >70 µm)")
```



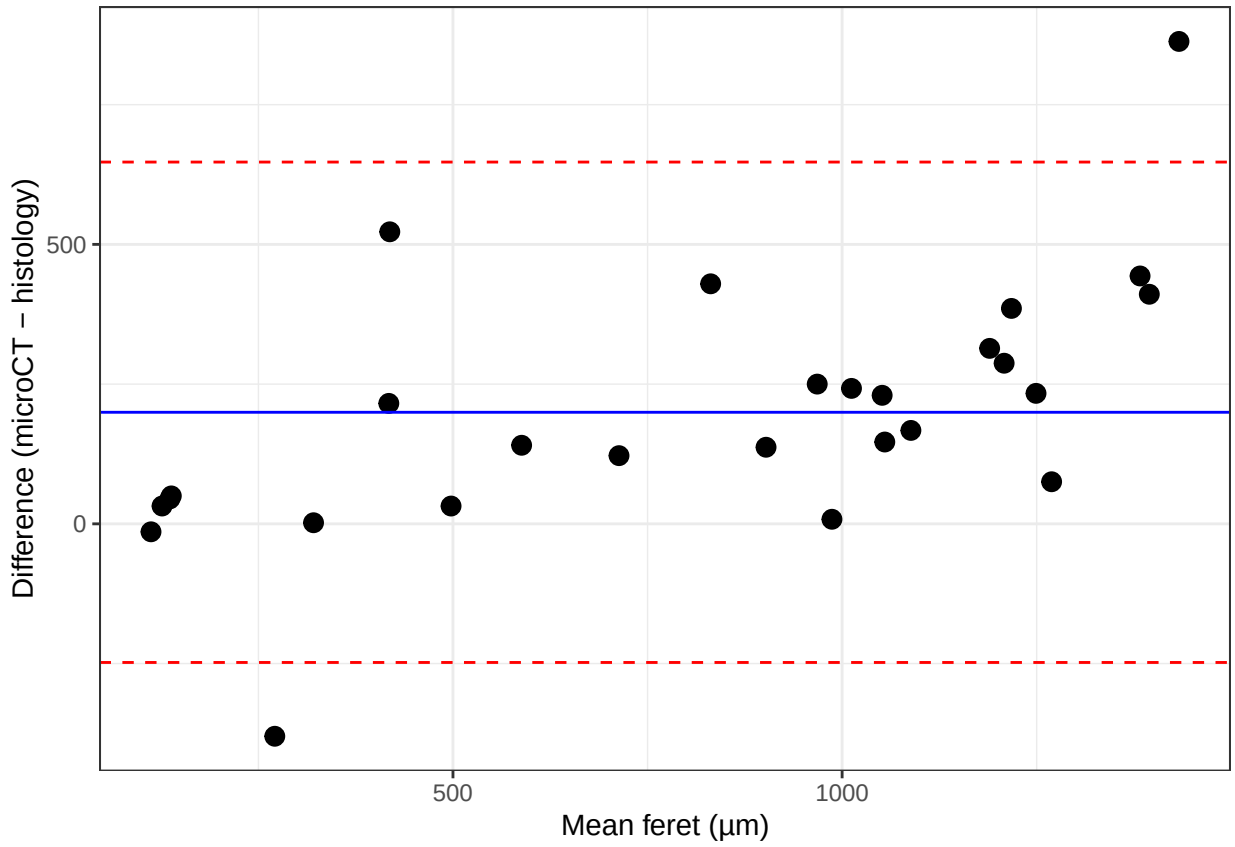

```
##########


# ---- 9a. Filter aggregated means ----
common <- specimen_stage %>%
  filter(mean_feret > 70) %>%
  group_by(specimen_id, stage) %>%
  filter(n() == 2) %>%  # keep only stages present in both methods
  ungroup()

# ---- 9b. Bland-Altman (filtered) ----
paired_filtered <- common %>%
  dplyr::select(specimen_id, species, stage, method, mean_feret) %>%
  pivot_wider(names_from = method, values_from = mean_feret) %>%
  filter(!is.na(microCT) & !is.na(histology)) %>%
  mutate(diff = microCT - histology,
         mean = (microCT + histology) / 2)

ggplot(paired_filtered, aes(x = mean, y = diff)) +
  geom_point(size = 3) +
  geom_hline(yintercept = mean(paired_filtered$diff), linetype = "solid", color = "blue") +
  geom_hline(yintercept = mean(paired_filtered$diff) + 1.96 * sd(paired_filtered$diff),
             linetype = "dashed", color = "red") +
  geom_hline(yintercept = mean(paired_filtered$diff) - 1.96 * sd(paired_filtered$diff),
             linetype = "dashed", color = "red") +
  labs(x = "Mean feret (µm)", y = "Difference (microCT - histology)") +
```

```
  theme_bw()
```




```
prop_bias_filt <- lm(diff ~ mean, data = paired_filtered)
summary(prop_bias_filt) # similar to unfiltered data, so still true (signficant)
```

```
##
## Call:
## lm(formula = diff ~ mean, data = paired_filtered)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -411.63  -79.86   -1.90   57.17  471.96
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -52.31885   76.27902  -0.686 0.499096
## mean          0.30951    0.08269   3.743 0.000955 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 186.5 on 25 degrees of freedom
## Multiple R-squared:  0.3592, Adjusted R-squared:  0.3335
## F-statistic: 14.01 on 1 and 25 DF,  p-value: 0.0009554
```

```r
# ---- 9c. Mixed-effects model (filtered) ----
# This is a sensitivity analysis using specimen-stage means (not individual gametes).
# The stage interaction is not significant, confirming that size-filtering removes
# the stage dependence observed in the unfiltered model.
m2 <- lmer(mean_feret ~ method * stage + (1 | specimen_id),
           data = common)
summary(m2)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: mean_feret ~ method * stage + (1 | specimen_id)
##    Data: common
##
## REML criterion at convergence: 582.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.04809 -0.43595  0.03877  0.61602  1.70170
##
## Random effects:
##  Groups      Name        Variance Std.Dev.
##  specimen_id (Intercept)  3941     62.78
##  Residual                37818    194.47
## Number of obs: 54, groups:  specimen_id, 12
##
## Fixed effects:
##                                              Estimate Std. Error      df
## (Intercept)                                    897.91      90.81   42.00
## methodmicroCT                                  196.01     122.99   33.28
## stagelarva                                     194.03     134.52   40.69
## stagelate oocyte or early embryo               -17.44     121.67   41.07
## stageprelarva                                  236.21     167.01   39.53
## stageprevitellogenic oocyte                   -790.58     146.21   40.36
## stagevitellogenic oocyte                      -546.29     116.17   38.70
## methodmicroCT:stagelarva                       183.08     184.49   33.28
## methodmicroCT:stagelate oocyte or early embryo  69.11     166.53   33.28
## methodmicroCT:stageprelarva                    142.48     230.10   33.28
## methodmicroCT:stageprevitellogenic oocyte     -153.84     200.85   33.28
## methodmicroCT:stagevitellogenic oocyte        -124.64     161.04   33.28
##                                              t value Pr(>|t|)
## (Intercept)                                    9.888 1.57e-12 ***
## methodmicroCT                                  1.594    0.120
## stagelarva                                     1.442    0.157
## stagelate oocyte or early embryo              -0.143    0.887
## stageprelarva                                  1.414    0.165
## stageprevitellogenic oocyte                   -5.407 3.13e-06 ***
## stagevitellogenic oocyte                      -4.703 3.23e-05 ***
## methodmicroCT:stagelarva                       0.992    0.328
## methodmicroCT:stagelate oocyte or early embryo 0.415    0.681
## methodmicroCT:stageprelarva                    0.619    0.540
## methodmicroCT:stageprevitellogenic oocyte     -0.766    0.449
## methodmicroCT:stagevitellogenic oocyte        -0.774    0.444
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##                 (Intr) mthdCT stglrv stooee stgprl stgpro stgvto mthdmcrCT:stgl
## methodmcrCT     -0.677
## stagelarva      -0.657  0.457
## stgltocyoee     -0.731  0.505  0.501
## stageprelrv     -0.518  0.368  0.344  0.394
## stgprvtllgo     -0.600  0.421  0.411  0.441  0.314
## stgvtllgnco     -0.750  0.529  0.511  0.563  0.414  0.480
## mthdmcrCT:stgl   0.451 -0.667 -0.686 -0.337 -0.245 -0.280 -0.353
## mthdCT:ooee      0.500 -0.739 -0.338 -0.684 -0.272 -0.311 -0.391  0.492
## mthdmcrCT:stgp   0.362 -0.535 -0.244 -0.270 -0.689 -0.225 -0.283  0.356
## mthdmcrCT:stgpo  0.415 -0.612 -0.280 -0.310 -0.225 -0.687 -0.324  0.408
## mthdmcrCT:stgvo  0.517 -0.764 -0.349 -0.386 -0.281 -0.321 -0.693  0.509
##                 mCooee mthdmcrCT:stgp mthdmcrCT:stgpo
## methodmcrCT
## stagelarva
## stgltocyoee
## stageprelrv
## stgprvtllgo
## stgvtllgnco
## mthdmcrCT:stgl
## mthdCT:ooee
## mthdmcrCT:stgp   0.395
## mthdmcrCT:stgpo  0.452  0.327
## mthdmcrCT:stgvo  0.564  0.408         0.468
```

```
anova(m2, type = 3)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##               Sum Sq Mean Sq NumDF  DenDF F value    Pr(>F)
## method        524165  524165     1 33.280  13.860 0.0007268 ***
## stage        7798769 1559754     5 41.322  41.244 5.141e-15 ***
## method:stage  191366   38273     5 33.280   1.012 0.4261231
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# ---- 9d. Weighted version (by number of gametes per mean) ----
# Weight by number of gametes per mean; results are consistent with unweighted model.

m2_weighted <- lmer(mean_feret ~ method * stage + (1 | specimen_id),
                    data = common, weights = n)
summary(m2_weighted)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: mean_feret ~ method * stage + (1 | specimen_id)
##    Data: common
## Weights: n
##
## REML criterion at convergence: 572.3
##
```

```
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.2277 -0.5165 -0.1185  0.3695  2.3954
##
## Random effects:
##  Groups      Name        Variance Std.Dev.
##  specimen_id (Intercept)      0     0.0
##  Residual                358949   599.1
## Number of obs: 54, groups:  specimen_id, 12
##
## Fixed effects:
##                                              Estimate Std. Error      df
## (Intercept)                                    970.86      93.57   42.00
## methodmicroCT                                  249.25     120.60   42.00
## stagelarva                                     149.31     203.44   42.00
## stagelate oocyte or early embryo               -36.64     115.83   42.00
## stageprelarva                                  185.65     152.03   42.00
## stageprevitellogenic oocyte                   -859.70      97.70   42.00
## stagevitellogenic oocyte                      -627.89     109.63   42.00
## methodmicroCT:stagelarva                        59.75     273.47   42.00
## methodmicroCT:stagelate oocyte or early embryo  -23.01     147.13   42.00
## methodmicroCT:stageprelarva                    164.74     197.88   42.00
## methodmicroCT:stageprevitellogenic oocyte      -213.93     125.50   42.00
## methodmicroCT:stagevitellogenic oocyte         -209.08     141.96   42.00
##                                              t value Pr(>|t|)
## (Intercept)                                   10.376 3.68e-13 ***
## methodmicroCT                                  2.067   0.0450 *
## stagelarva                                     0.734   0.4671
## stagelate oocyte or early embryo              -0.316   0.7533
## stageprelarva                                  1.221   0.2288
## stageprevitellogenic oocyte                   -8.799 4.42e-11 ***
## stagevitellogenic oocyte                      -5.727 9.79e-07 ***
## methodmicroCT:stagelarva                       0.218   0.8281
## methodmicroCT:stagelate oocyte or early embryo -0.156   0.8765
## methodmicroCT:stageprelarva                    0.833   0.4098
## methodmicroCT:stageprevitellogenic oocyte      -1.705   0.0957 .
## methodmicroCT:stagevitellogenic oocyte         -1.473   0.1483
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##                 (Intr) mthdCT stglrv stooee stgprl stgpro stgvto mthdmcrCT:stgl
## methodmcrCT     -0.776
## stagelarva      -0.460  0.357
## stgltocyoee     -0.808  0.627  0.372
## stageprelrv     -0.615  0.478  0.283  0.497
## stgprvtllgo     -0.958  0.743  0.440  0.774  0.589
## stgvtllgnco     -0.854  0.662  0.393  0.689  0.525  0.817
## mthdmcrCT:stgl   0.342 -0.441 -0.744 -0.276 -0.211 -0.328 -0.292
## mthdCT:ooee      0.636 -0.820 -0.292 -0.787 -0.391 -0.609 -0.543  0.361
## mthdmcrCT:stgp   0.473 -0.609 -0.217 -0.382 -0.768 -0.453 -0.404  0.269
## mthdmcrCT:stgpo  0.746 -0.961 -0.343 -0.602 -0.459 -0.778 -0.636  0.424
## mthdmcrCT:stgvo  0.659 -0.850 -0.303 -0.532 -0.406 -0.631 -0.772  0.375
##                 mCooee mthdmcrCT:stgp mthdmcrCT:stgpo
```

```
## methodmcrCT
## stagelarva
## stgltocyoee
## stageprelrv
## stgprvtllgo
## stgvtllgnco
## mthdmcrCT:stgl
## mthdCT:ooee
## mthdmcrCT:stgp   0.500
## mthdmcrCT:stgpo  0.788  0.586
## mthdmcrCT:stgvo  0.696  0.518          0.816
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

```
# ---- 9e. Effect sizes and percentage bias ----

# We are using emmeans, because this uses the model to predict what the mean would be for each stage-me


common <- common %>%
  mutate(
    stage_combined = case_when(
      stage %in% c("vitellogenic oocyte", "late oocyte or early embryo") ~ "Vitellogenic / late oocyte"
      stage %in% c("prelarva", "larva") ~ "Larva / prelarva",
      TRUE ~ as.character(stage)
    ),
    stage_combined = factor(stage_combined,
                            levels = c("previtellogenic oocyte",
                                       "Vitellogenic / late oocyte",
                                       "embryo",
                                       "Larva / prelarva"))
  )
#reaggregate because somehow I got duplicates above...
common <- common %>%
  group_by(specimen_id, species, stage_combined, method) %>%
  summarise(
    mean_feret = mean(mean_feret),
    n = sum(n),
    .groups = "drop"
  ) %>%
  group_by(specimen_id, stage_combined) %>%
  filter(n() == 2) %>%     # keep only stage-specimen combos in both methods
  ungroup()

# Sanity check: should return zero rows
common %>%
  dplyr::count(specimen_id, stage_combined, method) %>%
  filter(n > 1)
```

```
## # A tibble: 0 x 4
## # i 4 variables: specimen_id <chr>, stage_combined <fct>, method <chr>, n <int>
```

```r
m2_combined <- lmer(mean_feret ~ method * stage_combined + (1 | specimen_id),
                    data = common)
anova(m2_combined, type = 3)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                         Sum Sq Mean Sq NumDF  DenDF F value    Pr(>F)
## method                  379344  379344     1 29.902  7.9361  0.008503 **
## stage_combined         4884439 1628146     3 39.390 34.0617 4.994e-11 ***
## method:stage_combined   123641   41214     3 29.902  0.8622  0.471426
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
summary(m2_combined)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: mean_feret ~ method * stage_combined + (1 | specimen_id)
##    Data: common
##
## REML criterion at convergence: 591.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.03301 -0.54252  0.06308  0.57117  1.71078
##
## Random effects:
##  Groups      Name        Variance Std.Dev.
##  specimen_id (Intercept)  9975     99.88
##  Residual                47800    218.63
## Number of obs: 50, groups:  specimen_id, 12
##
## Fixed effects:
##                                                    Estimate Std. Error
## (Intercept)                                          181.00     135.08
## methodmicroCT                                         42.17     178.51
## stage_combinedVitellogenic / late oocyte             418.24     147.33
## stage_combinedembryo                                 737.16     167.08
## stage_combinedLarva / prelarva                       923.39     161.83
## methodmicroCT:stage_combinedVitellogenic / late oocyte   128.74     201.39
## methodmicroCT:stage_combinedembryo                   153.84     225.80
## methodmicroCT:stage_combinedLarva / prelarva         323.39     218.63
##                                                      df t value Pr(>|t|)
## (Intercept)                                       41.61   1.340  0.18755
## methodmicroCT                                     29.90   0.236  0.81488
## stage_combinedVitellogenic / late oocyte          35.91   2.839  0.00741
## stage_combinedembryo                              37.91   4.412 8.20e-05
## stage_combinedLarva / prelarva                    37.67   5.706 1.48e-06
## methodmicroCT:stage_combinedVitellogenic / late oocyte  29.90   0.639  0.52753
## methodmicroCT:stage_combinedembryo                29.90   0.681  0.50093
## methodmicroCT:stage_combinedLarva / prelarva      29.90   1.479  0.14956
##
## (Intercept)
```

```
## methodmicroCT
## stage_combinedVitellogenic / late oocyte               **
## stage_combinedembryo                                   ***
## stage_combinedLarva / prelarva                         ***
## methodmicroCT:stage_combinedVitellogenic / late oocyte
## methodmicroCT:stage_combinedembryo
## methodmicroCT:stage_combinedLarva / prelarva
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) mthdCT s_V/lo stg_cm st_L/p mCT/lo mtCT:_
## methodmcrCT -0.661
## stg_cmbV/lo -0.872  0.606
## stg_cmbndmb -0.776  0.534  0.706
## stg_cmbnL/p -0.803  0.552  0.737  0.649
## mthCT:_V/lo  0.586 -0.886 -0.683 -0.474 -0.489
## mthdmcrCT:_  0.522 -0.791 -0.479 -0.676 -0.436  0.701
## mthdCT:_L/p  0.540 -0.816 -0.495 -0.436 -0.676  0.724  0.645
```

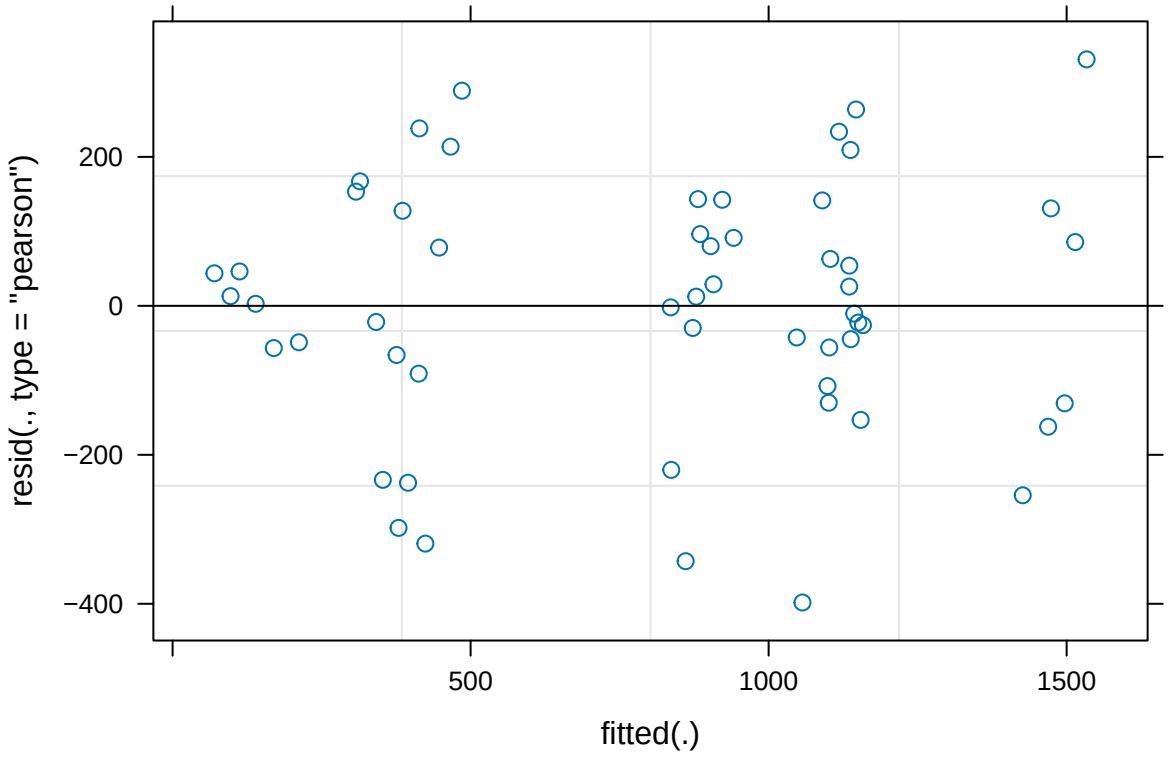
```
emm_m2c <- emmeans(m2_combined, ~ method | stage_combined)  # Computes the estimated marginal means (mo
contrasts_m2c <- contrast(emm_m2c, "pairwise", by = "stage_combined") #Compares the two methods (histol
summary(contrasts_m2c, by = NULL, adjust = "holm")
```

```
##  contrast            stage_combined             estimate    SE   df t.ratio
##  histology - microCT previtellogenic oocyte        -42.2 179.0 31.6  -0.236
##  histology - microCT Vitellogenic / late oocyte   -170.9  93.2 31.6  -1.833
##  histology - microCT embryo                       -196.0 138.0 31.6  -1.418
##  histology - microCT Larva / prelarva             -365.6 126.0 31.6  -2.896
##  p.value
##   0.8148
##   0.2286
##   0.3322
##   0.0272
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: holm method for 4 tests
```

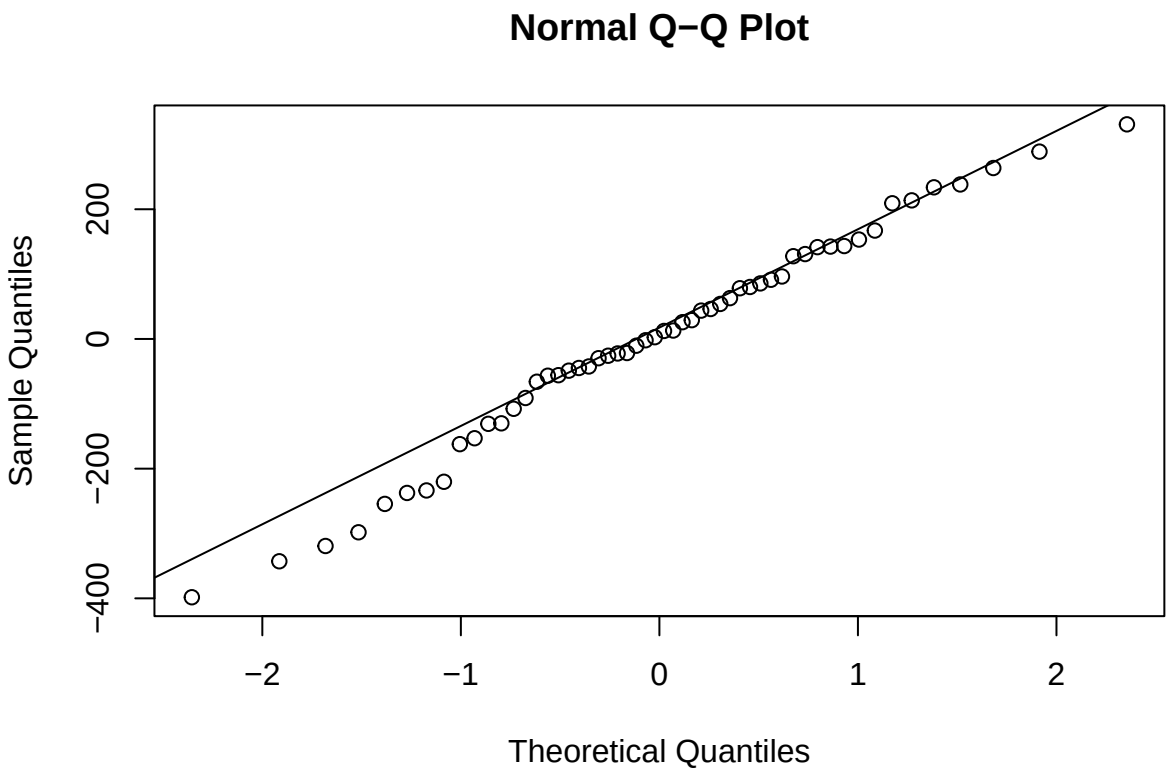
```
hist_means <- as.data.frame(emm_m2c) %>%
  filter(method == "histology") %>%
  dplyr::select(stage_combined, hist_mean = emmean)

#

# ---- 9f. Model diagnostics for filtered model ----
plot(m2)          # residuals vs fitted
```




```
qqnorm(resid(m2))
qqline(resid(m2))
```




```
# On combined stages:

# Histology emmeans
hist_em <- as.data.frame(emm_combined) %>%
  filter(method == "histology") %>%
  dplyr::select(stage_combined, hist_mean = emmean)

# Contrasts (histology - microCT)
contr <- as.data.frame(contrasts_combined) %>%
  mutate(microCT_minus_histology = -estimate)  # positive = microCT larger

# Join and compute percent bias
pct_table <- left_join(contr, hist_em, by = "stage_combined") %>%
  mutate(percent_bias = (microCT_minus_histology / hist_mean) * 100) %>%
  dplyr::select(stage_combined, contrast_est = estimate, hist_mean,
                microCT_minus_histology, percent_bias)
pct_table
```

```
##               stage_combined contrast_est hist_mean microCT_minus_histology
## 1                     embryo   -274.61255  999.5408               274.61255
## 2          Larva / prelarva   -449.99112 1017.9186               449.99112
## 3    previtellogenic oocyte    -23.34775  437.3060                23.34775
## 4 Vitellogenic / late oocyte   -177.57135  615.0479               177.57135
##   percent_bias
## 1    27.473869
## 2    44.206986
```

```
## 3     5.338997
## 4    28.871140
```

```
# Overall marginal means averaged over stages
emm_all <- emmeans(model_combined, ~ method)
emm_tab <- as.data.frame(emm_all)
(overall_pct <- (emm_tab$emmean[2] - emm_tab$emmean[1]) / emm_tab$emmean[1] * 100)
```

```
## [1] 30.14915
```

```
pairs(emm_all)
```

```
##  contrast            estimate SE   df t.ratio p.value
##  histology - microCT     -231 14 1886 -16.562 <0.0001
##
## Results are averaged over the levels of: stage_combined
## Degrees-of-freedom method: kenward-roger
```

---

10. SPECIES AND PHYLUM EFFECTS

```
# ---- 10a. Species as a fixed effect ----
# Check how the species differed, so adding species into the linear mixed effects model:

model_species <- lmer(feret_micron ~ method * species + stage + (1 | specimen_id),
                      data = gametes)
summary(model_species)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: feret_micron ~ method * species + stage + (1 | specimen_id)
##    Data: gametes
##
## REML criterion at convergence: 25769.2
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -5.2418 -0.3869 -0.0583  0.3217  8.2453
##
## Random effects:
##  Groups      Name        Variance Std.Dev.
##  specimen_id (Intercept)  2282     47.78
##  Residual                16921    130.08
## Number of obs: 2056, groups:  specimen_id, 12
##
## Fixed effects:
##                                      Estimate Std. Error       df t value
## (Intercept)                            807.72      38.24    21.25  21.121
```

```
## methodmicroCT                                205.50      24.02  2038.04   8.556
## speciesParagorgia arborea                    330.23      46.26    11.43   7.139
## speciesStelodoryx oxeata                     152.90      47.48    12.68   3.221
## speciesTedania kagalaskai                    165.54      45.17    10.40   3.665
## stagelarva                                   281.42      30.19  2028.41   9.320
## stagelate oocyte or early embryo              51.46      21.56  2042.38   2.386
## stageprelarva                                293.75      23.90  2031.22  12.289
## stageprevitellogenic oocyte                -1007.91      21.09  2022.10 -47.783
## stagevitellogenic oocyte                    -751.44      20.01  2005.63 -37.559
## methodmicroCT:speciesParagorgia arborea     -168.18      25.06  2037.91  -6.711
## methodmicroCT:speciesStelodoryx oxeata        88.31      32.51  2038.54   2.717
## methodmicroCT:speciesTedania kagalaskai      -53.63      28.85  2037.83  -1.859
##                                             Pr(>|t|)
## (Intercept)                                 9.60e-16 ***
## methodmicroCT                                < 2e-16 ***
## speciesParagorgia arborea                   1.54e-05 ***
## speciesStelodoryx oxeata                     0.00689 **
## speciesTedania kagalaskai                    0.00408 **
## stagelarva                                   < 2e-16 ***
## stagelate oocyte or early embryo             0.01711 *
## stageprelarva                                < 2e-16 ***
## stageprevitellogenic oocyte                  < 2e-16 ***
## stagevitellogenic oocyte                     < 2e-16 ***
## methodmicroCT:speciesParagorgia arborea 2.49e-11 ***
## methodmicroCT:speciesStelodoryx oxeata       0.00665 **
## methodmicroCT:speciesTedania kagalaskai      0.06320 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(model_species, type = 3)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                  Sum Sq  Mean Sq NumDF   DenDF  F value    Pr(>F)
## method          5613857  5613857     1 2041.33  331.773 < 2.2e-16 ***
## species          571389   190463     3    8.54   11.256  0.002488 **
## stage          89821811 17964362     5 2021.27 1061.675 < 2.2e-16 ***
## method:species  2906234   968745     3 2039.27   57.252 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# ---- Pairwise contrasts per species ----
# Use unweighted emmeans for contrasts (the differences are the same)
emm_species <- emmeans(model_species, ~ method | species)
contrasts_species <- contrast(emm_species, "pairwise", by = "species", adjust = "tukey")
print(contrasts_species)
```

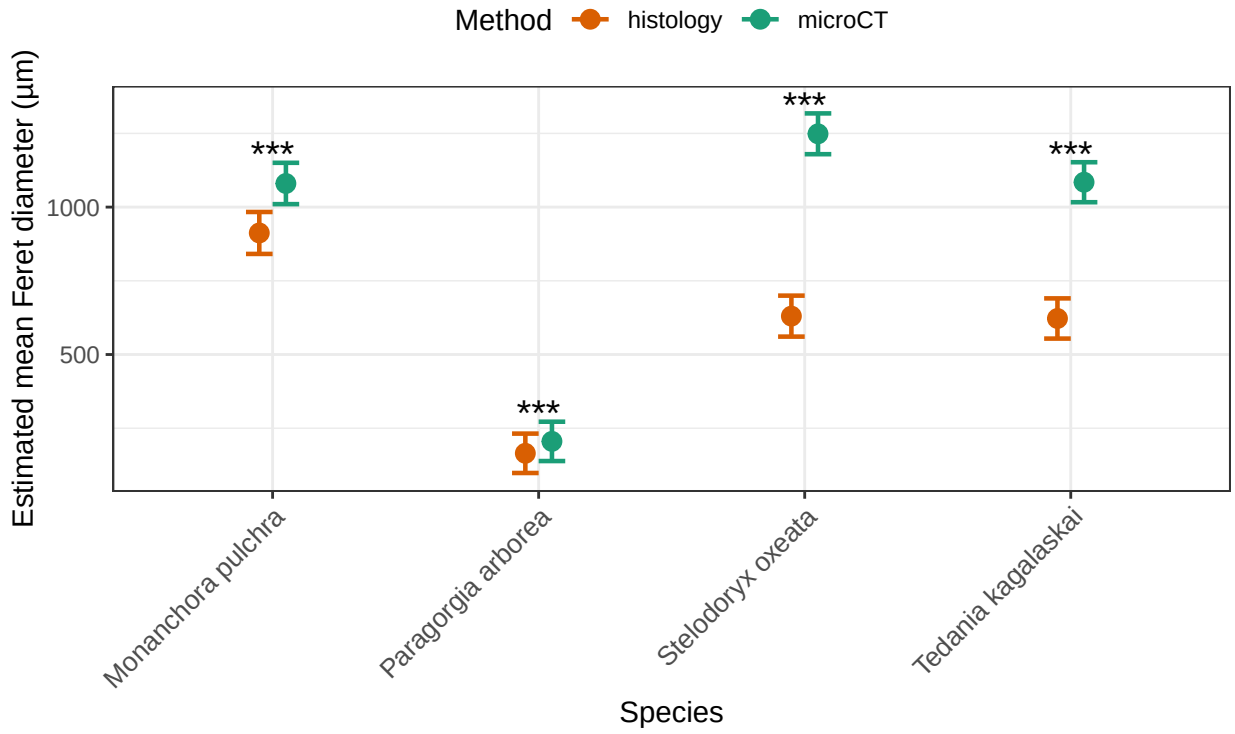
```
## species = Monanchora pulchra:
##  contrast            estimate    SE   df t.ratio p.value
##  histology - microCT   -205.5 24.00 2038  -8.554 <0.0001
##
## species = Paragorgia arborea:
##  contrast            estimate    SE   df t.ratio p.value
##  histology - microCT    -37.3  7.16 2037  -5.214 <0.0001
```

```
##
## species = Stelodoryx oxeata:
##  contrast            estimate    SE   df t.ratio p.value
##  histology - microCT   -293.8 22.00 2042 -13.359 <0.0001
##
## species = Tedania kagalaskai:
##  contrast            estimate    SE   df t.ratio p.value
##  histology - microCT   -151.9 16.40 2039  -9.271 <0.0001
##
## Results are averaged over the levels of: stage
## Degrees-of-freedom method: kenward-roger
```

```
# ---- Species-specific means with weighted emmeans ----
# Use weights = "cells" to average only over stages that exist in each species.
# This prevents the model from extrapolating to stages that are absent (e.g.,
# prelarva and late oocyte/early embryo in Paragorgia).
emm_species_weighted <- emmeans(model_species, ~ method | species, weights = "cells")
emm_species_df <- as.data.frame(emm_species_weighted)

# ---- Annotation: significance stars ----
# All pairwise contrasts are significant (p < 0.001), so we place "***" above each species.
annotation_species <- emm_species_df %>%
  group_by(species) %>%
  summarise(
    y_star = max(upper.CL) + 30,  # 30 µm above the highest error bar
    .groups = "drop"
  )

# ---- Publication-ready species plot ----
ggplot(emm_species_df, aes(x = species, y = emmean, color = method)) +
  geom_point(position = position_dodge(0.2), size = 3) +
  geom_errorbar(aes(ymin = lower.CL, ymax = upper.CL),
                position = position_dodge(0.2), width = 0.2, linewidth = 0.8) +
  # Significance stars
  geom_text(data = annotation_species,
            aes(x = species, y = y_star, label = "***"),
            color = "black", size = 5, inherit.aes = FALSE) +
  scale_color_manual(values = c("histology" = "#d95f02", "microCT" = "#1b9e77"),
                     name = "Method") +
  theme_bw() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1, size = 10),
    legend.position = "top"
  ) +
  labs(
    y = "Estimated mean Feret diameter (µm)",
    x = "Species",
    caption = "*** p < 0.001 for histology vs. microCT within each species (Tukey-adjusted).\nMeans are
  )
```



*** p < 0.001 for histology vs. microCT within each species (Tukey−adjusted).
Means are weighted estimated marginal means (emmeans) adjusted for developmental stage, weighted by the observed stage distribution per species (stages absent from a species are excluded).

```
# ---- 10b. Phylum effects ----
model_group <- lmer(feret_micron ~ method * phylum * stage + (1 | specimen_id),
                    data = gametes)
summary(model_group)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: feret_micron ~ method * phylum * stage + (1 | specimen_id)
##    Data: gametes
##
## REML criterion at convergence: 25319.8
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.8317 -0.4279 -0.0813  0.3249  7.8695
##
## Random effects:
##  Groups      Name        Variance Std.Dev.
##  specimen_id (Intercept)  5284     72.69
##  Residual                13957    118.14
## Number of obs: 2056, groups:  specimen_id, 12
##
## Fixed effects:
##                                                  Estimate Std. Error
## (Intercept)                                        511.96      80.30
## methodmicroCT                                      140.56      90.23
```

```
## phylumPorifera                                          472.68      86.43
## stagelarva                                              465.30     108.60
## stagelate oocyte or early embryo                        -40.87      27.18
## stageprelarva                                            96.47      30.22
## stageprevitellogenic oocyte                            -382.89      68.73
## stagevitellogenic oocyte                                -52.01      70.52
## methodmicroCT:phylumPorifera                            115.56      93.46
## methodmicroCT:stagelarva                                722.56     148.66
## methodmicroCT:stagelate oocyte or early embryo          -50.75      29.59
## methodmicroCT:stageprelarva                             163.45      37.09
## methodmicroCT:stageprevitellogenic oocyte              -103.20      90.49
## methodmicroCT:stagevitellogenic oocyte                 -158.17      92.51
## phylumPorifera:stagelarva                              -265.32     116.76
## phylumPorifera:stageprevitellogenic oocyte             -636.66      75.61
## phylumPorifera:stagevitellogenic oocyte                -742.44      74.94
## methodmicroCT:phylumPorifera:stagelarva                -728.48     158.04
## methodmicroCT:phylumPorifera:stagevitellogenic oocyte  -166.23      99.21
##                                                            df t value Pr(>|t|)
## (Intercept)                                             110.71   6.375 4.33e-09
## methodmicroCT                                          2026.37   1.558  0.11944
## phylumPorifera                                           83.80   5.469 4.60e-07
## stagelarva                                             2027.37   4.285 1.92e-05
## stagelate oocyte or early embryo                       1813.26  -1.503  0.13289
## stageprelarva                                          1992.81   3.193  0.00143
## stageprevitellogenic oocyte                            2026.67  -5.571 2.87e-08
## stagevitellogenic oocyte                               2026.61  -0.738  0.46090
## methodmicroCT:phylumPorifera                           2026.48   1.236  0.21645
## methodmicroCT:stagelarva                               2026.37   4.861 1.26e-06
## methodmicroCT:stagelate oocyte or early embryo         2029.96  -1.715  0.08648
## methodmicroCT:stageprelarva                            2029.36   4.407 1.10e-05
## methodmicroCT:stageprevitellogenic oocyte              2026.37  -1.140  0.25427
## methodmicroCT:stagevitellogenic oocyte                 2026.40  -1.710  0.08745
## phylumPorifera:stagelarva                              2032.81  -2.272  0.02317
## phylumPorifera:stageprevitellogenic oocyte             2034.26  -8.421  < 2e-16
## phylumPorifera:stagevitellogenic oocyte                2030.43  -9.907  < 2e-16
## methodmicroCT:phylumPorifera:stagelarva                2026.65  -4.609 4.29e-06
## methodmicroCT:phylumPorifera:stagevitellogenic oocyte  2028.12  -1.675  0.09399
##
## (Intercept)                                            ***
## methodmicroCT
## phylumPorifera                                         ***
## stagelarva                                             ***
## stagelate oocyte or early embryo
## stageprelarva                                          **
## stageprevitellogenic oocyte                            ***
## stagevitellogenic oocyte
## methodmicroCT:phylumPorifera
## methodmicroCT:stagelarva                               ***
## methodmicroCT:stagelate oocyte or early embryo         .
## methodmicroCT:stageprelarva                            ***
## methodmicroCT:stageprevitellogenic oocyte
## methodmicroCT:stagevitellogenic oocyte                 .
## phylumPorifera:stagelarva                              *
## phylumPorifera:stageprevitellogenic oocyte             ***
```

```
## phylumPorifera:stagevitellogenic oocyte               ***
## methodmicroCT:phylumPorifera:stagelarva               ***
## methodmicroCT:phylumPorifera:stagevitellogenic oocyte .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## fit warnings:
## fixed-effect model matrix is rank deficient so dropping 5 columns / coefficients
```

```
anova(model_group)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                        Sum Sq Mean Sq NumDF   DenDF  F value    Pr(>F)
## method                2409036 2409036     1 2027.39 172.6013 < 2.2e-16 ***
## phylum                    276     276     1   12.56   0.0198 0.8904117
## stage                 43851529 8770306     5 2017.95 628.3701 < 2.2e-16 ***
## method:phylum           161017  161017     1 2026.67  11.5365 0.0006956 ***
## method:stage           2915346  583069     5 2030.62  41.7754 < 2.2e-16 ***
## phylum:stage           3729925 1243308     3 2033.56  89.0799 < 2.2e-16 ***
## method:phylum:stage     312341  156170     2 2028.14  11.1892  1.47e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
model_phylum <- lmer(feret_micron ~ method * phylum + (1 | specimen_id),
                     data = gametes)
summary(model_phylum)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: feret_micron ~ method * phylum + (1 | specimen_id)
##    Data: gametes
##
## REML criterion at convergence: 28522.8
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.1496 -0.3103 -0.0872  0.2864  6.8457
##
## Random effects:
##  Groups      Name        Variance Std.Dev.
##  specimen_id (Intercept)  9969     99.84
##  Residual                62171    249.34
## Number of obs: 2056, groups:  specimen_id, 12
##
## Fixed effects:
##                              Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)                   185.318     58.976    8.961   3.142  0.01195 *
## methodmicroCT                  42.166     13.708 2043.869   3.076  0.00213 **
## phylumPorifera                525.717     69.656    9.798   7.547 2.19e-05 ***
## methodmicroCT:phylumPorifera  392.344     25.262 2045.583  15.531  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Correlation of Fixed Effects:
##             (Intr) mthdCT phylmP
## methodmcrCT -0.144
## phylumPorfr -0.847  0.122
## mthdmcrCT:P  0.078 -0.543 -0.204
```

```
anova(model_phylum)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                 Sum Sq  Mean Sq NumDF   DenDF F value    Pr(>F)
## method        22135603 22135603     1 2045.58  356.04 < 2.2e-16 ***
## phylum         6962972  6962972     1    9.01  112.00 2.205e-06 ***
## method:phylum 14996120 14996120     1 2045.58  241.21 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

###10.2 SAME AS ABOVE BUT FILTERED

```
# ---- 10a. Species as a fixed effect (FILTERED DATA >70 µm) ----
# Use gametes_large (already filtered to >70 µm)
model_species_filtered <- lmer(feret_micron ~ method * species + stage + (1 | specimen_id),
                               data = gametes_large)
summary(model_species_filtered)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: feret_micron ~ method * species + stage + (1 | specimen_id)
##    Data: gametes_large
##
## REML criterion at convergence: 23923.9
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -5.2031 -0.3622 -0.0638  0.3088  8.1163
##
## Random effects:
##  Groups      Name        Variance Std.Dev.
##  specimen_id (Intercept)  2426     49.25
##  Residual                17535    132.42
## Number of obs: 1904, groups:  specimen_id, 12
##
## Fixed effects:
##                                          Estimate Std. Error       df t value
## (Intercept)                                801.84      39.53    21.77  20.287
## methodmicroCT                              209.19      24.59  1886.78   8.509
## speciesParagorgia arborea                  344.43      48.70    12.60   7.072
## speciesStelodoryx oxeata                   165.33      49.37    13.31   3.349
## speciesTedania kagalaskai                  175.65      46.76    10.72   3.756
## stagelarva                                 279.63      30.84  1881.10   9.067
## stagelate oocyte or early embryo            54.09      22.36  1890.91   2.420
## stageprelarva                              295.36      24.98  1872.59  11.826
```

```
## stageprevitellogenic oocyte             -1000.33      22.90  1882.70 -43.681
## stagevitellogenic oocyte                 -737.21      21.04  1873.26 -35.035
## methodmicroCT:speciesParagorgia arborea  -185.80      25.80  1886.62  -7.201
## methodmicroCT:speciesStelodoryx oxeata     77.88      33.49  1886.09   2.326
## methodmicroCT:speciesTedania kagalaskai   -66.12      30.10  1885.65  -2.196
##                                         Pr(>|t|)
## (Intercept)                             1.26e-15 ***
## methodmicroCT                            < 2e-16 ***
## speciesParagorgia arborea               9.94e-06 ***
## speciesStelodoryx oxeata                 0.00508 **
## speciesTedania kagalaskai                0.00332 **
## stagelarva                               < 2e-16 ***
## stagelate oocyte or early embryo         0.01563 *
## stageprelarva                            < 2e-16 ***
## stageprevitellogenic oocyte              < 2e-16 ***
## stagevitellogenic oocyte                 < 2e-16 ***
## methodmicroCT:speciesParagorgia arborea 8.59e-13 ***
## methodmicroCT:speciesStelodoryx oxeata   0.02014 *
## methodmicroCT:speciesTedania kagalaskai  0.02820 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(model_species_filtered, type = 3)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                   Sum Sq  Mean Sq NumDF   DenDF F value    Pr(>F)
## method          5047501  5047501     1 1889.20 287.856 < 2.2e-16 ***
## species          568834   189611     3    9.11  10.813  0.002347 **
## stage          65868136 13173627     5 1874.86 751.284 < 2.2e-16 ***
## method:species  3025380  1008460     3 1886.66  57.512 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# ---- Pairwise contrasts per species (filtered) ----
emm_species_filtered <- emmeans(model_species_filtered, ~ method | species)
contrasts_species_filtered <- contrast(emm_species_filtered, "pairwise", by = "species", adjust = "tukey
print(contrasts_species_filtered)
```

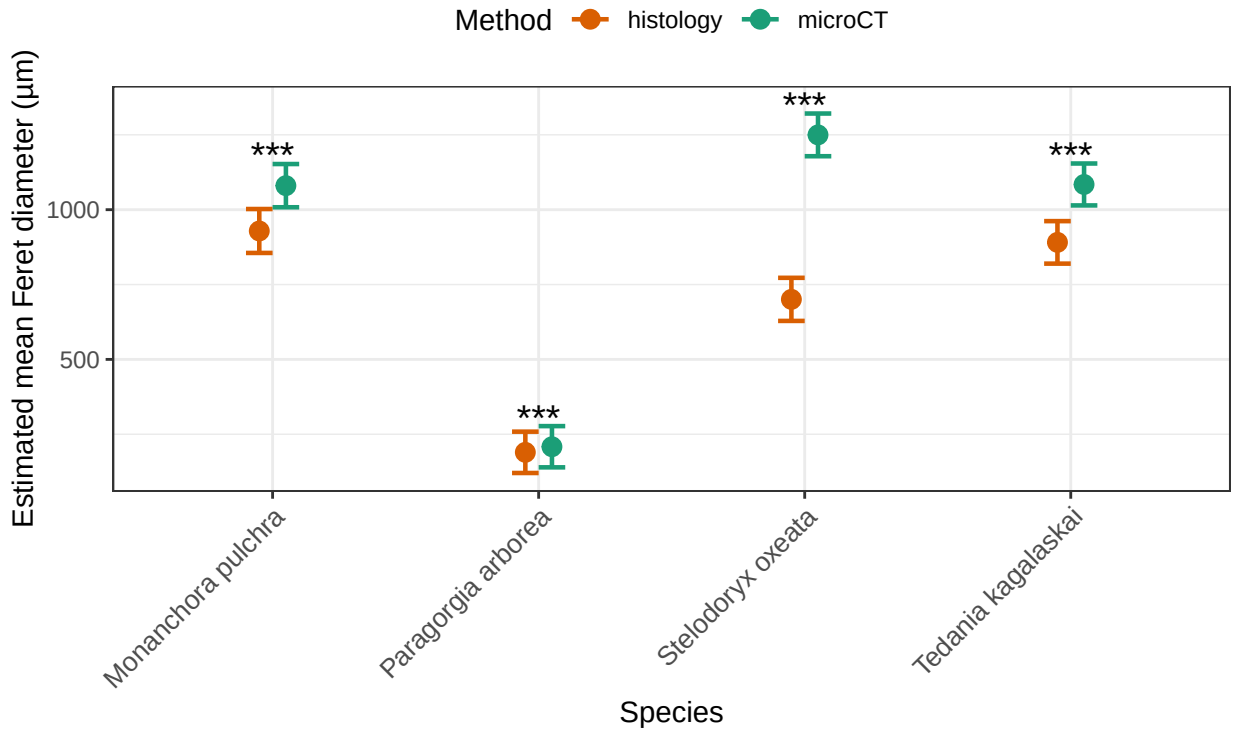
```
## species = Monanchora pulchra:
##  contrast            estimate    SE   df t.ratio p.value
##  histology - microCT   -209.2 24.60 1887  -8.506 <0.0001
##
## species = Paragorgia arborea:
##  contrast            estimate    SE   df t.ratio p.value
##  histology - microCT    -23.4  7.83 1885  -2.987  0.0029
##
## species = Stelodoryx oxeata:
##  contrast            estimate    SE   df t.ratio p.value
##  histology - microCT   -287.1 22.80 1888 -12.607 <0.0001
##
## species = Tedania kagalaskai:
##  contrast            estimate    SE   df t.ratio p.value
##  histology - microCT   -143.1 17.80 1888  -8.056 <0.0001
```

```
##
## Results are averaged over the levels of: stage
## Degrees-of-freedom method: kenward-roger
```

```r
# ---- Species-specific means with weighted emmeans (filtered) ----
# Use weights = "cells" to average only over stages that exist in each species.
emm_species_weighted_filtered <- emmeans(model_species_filtered, ~ method | species, weights = "cells")
emm_species_df_filtered <- as.data.frame(emm_species_weighted_filtered)

# ---- Annotation: significance stars ----
annotation_species_filtered <- emm_species_df_filtered %>%
  group_by(species) %>%
  summarise(
    y_star = max(upper.CL) + 30,
    .groups = "drop"
  )

# ---- Publication-ready species plot (filtered) ----
ggplot(emm_species_df_filtered, aes(x = species, y = emmean, color = method)) +
  geom_point(position = position_dodge(0.2), size = 3) +
  geom_errorbar(aes(ymin = lower.CL, ymax = upper.CL),
                position = position_dodge(0.2), width = 0.2, linewidth = 0.8) +
  geom_text(data = annotation_species_filtered,
            aes(x = species, y = y_star, label = "***"),
            color = "black", size = 5, inherit.aes = FALSE) +
  scale_color_manual(values = c("histology" = "#d95f02", "microCT" = "#1b9e77"),
                     name = "Method") +
  theme_bw() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1, size = 10),
    legend.position = "top"
  ) +
  labs(
    y = "Estimated mean Feret diameter (µm)",
    x = "Species",
    caption = "*** p < 0.001 for histology vs. microCT within each species (Tukey-adjusted).\nData resti
  )
```



*** p < 0.001 for histology vs. microCT within each species (Tukey−adjusted).
Data restricted to gametes >70 µm. Means are weighted estimated marginal means (emmeans)
adjusted for developmental stage, weighted by the observed stage distribution per species.

```
# ---- 10b. Phylum effects (filtered) ----
model_group_filtered <- lmer(feret_micron ~ method * phylum * stage + (1 | specimen_id),
                             data = gametes_large)
summary(model_group_filtered)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: feret_micron ~ method * phylum * stage + (1 | specimen_id)
##    Data: gametes_large
##
## REML criterion at convergence: 23467.7
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.9191 -0.4183 -0.0876  0.3281  7.7622
##
## Random effects:
##  Groups      Name        Variance Std.Dev.
##  specimen_id (Intercept)  5407     73.53
##  Residual                 14122    118.84
## Number of obs: 1904, groups:  specimen_id, 12
##
## Fixed effects:
##                                                  Estimate Std. Error
## (Intercept)                                        511.75      80.92
```

```
## methodmicroCT                                          140.56      90.76
## phylumPorifera                                         482.22      87.15
## stagelarva                                             461.22     109.32
## stagelate oocyte or early embryo                       -59.25      27.67
## stageprelarva                                           73.84      30.76
## stageprevitellogenic oocyte                           -363.30      69.23
## stagevitellogenic oocyte                               -51.07      70.95
## methodmicroCT:phylumPorifera                           117.34      94.01
## methodmicroCT:stagelarva                               722.56     149.53
## methodmicroCT:stagelate oocyte or early embryo         -48.69      29.78
## methodmicroCT:stageprelarva                            158.77      37.33
## methodmicroCT:stageprevitellogenic oocyte              -118.92      91.07
## methodmicroCT:stagevitellogenic oocyte                 -156.97      93.05
## phylumPorifera:stagelarva                              -278.59     117.57
## phylumPorifera:stagevitellogenic oocyte                -671.38      75.84
## methodmicroCT:phylumPorifera:stagelarva                -722.37     158.98
## methodmicroCT:phylumPorifera:stagevitellogenic oocyte  -265.01     100.39
##                                                            df t value Pr(>|t|)
## (Intercept)                                             110.79   6.324 5.52e-09
## methodmicroCT                                          1875.61   1.549  0.12163
## phylumPorifera                                           84.06   5.533 3.49e-07
## stagelarva                                             1876.71   4.219 2.57e-05
## stagelate oocyte or early embryo                       1649.20  -2.142  0.03238
## stageprelarva                                          1829.08   2.401  0.01646
## stageprevitellogenic oocyte                            1875.93  -5.248 1.71e-07
## stagevitellogenic oocyte                               1875.86  -0.720  0.47171
## methodmicroCT:phylumPorifera                           1875.72   1.248  0.21215
## methodmicroCT:stagelarva                               1875.61   4.832 1.46e-06
## methodmicroCT:stagelate oocyte or early embryo         1879.14  -1.635  0.10221
## methodmicroCT:stageprelarva                            1878.48   4.253 2.21e-05
## methodmicroCT:stageprevitellogenic oocyte              1875.61  -1.306  0.19180
## methodmicroCT:stagevitellogenic oocyte                 1875.64  -1.687  0.09178
## phylumPorifera:stagelarva                              1882.00  -2.370  0.01791
## phylumPorifera:stagevitellogenic oocyte                1879.12  -8.852  < 2e-16
## methodmicroCT:phylumPorifera:stagelarva                1875.90  -4.544 5.87e-06
## methodmicroCT:phylumPorifera:stagevitellogenic oocyte  1878.35  -2.640  0.00836
##
## (Intercept)                                            ***
## methodmicroCT
## phylumPorifera                                         ***
## stagelarva                                             ***
## stagelate oocyte or early embryo                       *
## stageprelarva                                          *
## stageprevitellogenic oocyte                            ***
## stagevitellogenic oocyte
## methodmicroCT:phylumPorifera
## methodmicroCT:stagelarva                               ***
## methodmicroCT:stagelate oocyte or early embryo
## methodmicroCT:stageprelarva                            ***
## methodmicroCT:stageprevitellogenic oocyte
## methodmicroCT:stagevitellogenic oocyte                 .
## phylumPorifera:stagelarva                              *
## phylumPorifera:stagevitellogenic oocyte                ***
## methodmicroCT:phylumPorifera:stagelarva                ***
```

```
## methodmicroCT:phylumPorifera:stagevitellogenic oocyte **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## fit warnings:
## fixed-effect model matrix is rank deficient so dropping 6 columns / coefficients
```

```
anova(model_group_filtered)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                        Sum Sq Mean Sq NumDF   DenDF  F value    Pr(>F)
## method                2221901 2221901     1 1876.46 157.3381 < 2.2e-16 ***
## phylum                  15952   15952     1   14.86   1.1296    0.3048
## stage                 28930752 5786150     5 1859.37 409.7311 < 2.2e-16 ***
## method:phylum           215696  215696     1 1875.94  15.2739 9.629e-05 ***
## method:stage           3496860  699372     5 1879.96  49.5242 < 2.2e-16 ***
## phylum:stage           3495961 1747981     2 1883.72 123.7787 < 2.2e-16 ***
## method:phylum:stage     291940  145970     2 1878.08  10.3365 3.431e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
model_phylum_filtered <- lmer(feret_micron ~ method * phylum + (1 | specimen_id),
                              data = gametes_large)
summary(model_phylum_filtered)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: feret_micron ~ method * phylum + (1 | specimen_id)
##    Data: gametes_large
##
## REML criterion at convergence: 26136
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.6931 -0.3040 -0.0931  0.3091  7.3126
##
## Random effects:
##  Groups      Name        Variance Std.Dev.
##  specimen_id (Intercept)  7034     83.87
##  Residual                53795    231.94
## Number of obs: 1904, groups:  specimen_id, 12
##
## Fixed effects:
##                              Estimate Std. Error       df t value Pr(>|t|)
## (Intercept)                   212.475     50.074    8.820   4.243  0.00227 **
## methodmicroCT                  19.960     13.716 1892.335   1.455  0.14579
## phylumPorifera                622.303     59.625    9.965  10.437  1.1e-06 ***
## methodmicroCT:phylumPorifera  266.967     25.047 1894.441  10.659  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
```

```
##             (Intr) mthdCT phylmP
## methodmcrCT -0.183
## phylumPorfr -0.840  0.153
## mthdmcrCT:P  0.100 -0.548 -0.252
```

```
anova(model_phylum_filtered)
```

```
## Type III Analysis of Variance Table with Satterthwaite's method
##                 Sum Sq Mean Sq NumDF   DenDF F value    Pr(>F)
## method        8075803 8075803     1 1894.44  150.12 < 2.2e-16 ***
## phylum        9212570 9212570     1    8.77  171.25  4.71e-07 ***
## method:phylum 6111473 6111473     1 1894.44  113.61 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Check the performances of the tests/data:
library(see)
performance::check_model(model_combined)
performance::check_model(model_species_filtered)
performance::check_model(model_phylum_filtered)


#Also do this to get the p-values:
# Adjust across all four stage-specific method contrasts
summary(contrasts_combined, by = NULL, adjust = "holm")
```

```
##  contrast            stage_combined             estimate   SE   df t.ratio
##  histology - microCT embryo                       -274.6 36.4 1885  -7.539
##  histology - microCT Larva / prelarva             -450.0 36.8 1887 -12.243
##  histology - microCT previtellogenic oocyte         -23.3 11.6 1885  -2.020
##  histology - microCT Vitellogenic / late oocyte   -177.6 17.7 1888 -10.007
##  p.value
##  <0.0001
##  <0.0001
##   0.0435
##  <0.0001
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: holm method for 4 tests
```

```
# Adjust across all four species-specific method contrasts
summary(contrasts_species_filtered, by = NULL, adjust = "holm")
```

```
##  contrast            species            estimate    SE   df t.ratio p.value
##  histology - microCT Monanchora pulchra   -209.2 24.60 1887  -8.506 <0.0001
##  histology - microCT Paragorgia arborea    -23.4  7.83 1885  -2.987  0.0029
##  histology - microCT Stelodoryx oxeata    -287.1 22.80 1888 -12.607 <0.0001
##  histology - microCT Tedania kagalaskai   -143.1 17.80 1888  -8.056 <0.0001
##
## Results are averaged over the levels of: stage
## Degrees-of-freedom method: kenward-roger
## P value adjustment: holm method for 4 tests
```

```r
# Phylum-specific contrasts
emm_phylum_filtered <- emmeans(
  model_phylum_filtered,
  ~ method | phylum
)

contrasts_phylum_filtered <- contrast(
  emm_phylum_filtered,
  "pairwise",
  by = "phylum"
)

summary(contrasts_phylum_filtered, by = NULL, adjust = "holm")
```

```
##  contrast            phylum   estimate   SE   df t.ratio p.value
##  histology - microCT Cnidaria      -20 13.7 1892  -1.455  0.1458
##  histology - microCT Porifera     -287 21.0 1895 -13.687 <0.0001
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: holm method for 2 tests
```

###10.3 INDIVIDUAL EFFECTS

```r
#Whats the effect of the specimen intercept?

library(RLRsim)

# 1. Overall and combined-stage contrasts
test_individual_combined <- exactRLRT(
    model_combined, nsim = 10000, seed = 101)

# 2. Species contrasts
test_individual_species <- exactRLRT(
     model_species_filtered, nsim = 10000, seed = 102)

# 3. Phylum contrasts
test_individual_phylum <- exactRLRT(
     model_phylum_filtered, nsim = 10000, seed = 103)

# Print the three tests
test_individual_combined
```

```
##
##  simulated finite sample distribution of RLRT.
##
##  (p-value based on 10000 simulated values)
##
## data:
## RLRT = 668.82, p-value < 2.2e-16
```

```r
test_individual_species
```

```
##
##  simulated finite sample distribution of RLRT.
##
##  (p-value based on 10000 simulated values)
##
## data:
## RLRT = 53.453, p-value < 2.2e-16
```

```
test_individual_phylum
```

```
##
##  simulated finite sample distribution of RLRT.
##
##  (p-value based on 10000 simulated values)
##
## data:
## RLRT = 93.282, p-value < 2.2e-16
```

```
### Also just to satsify anyone arguing with the model:
#Using RANOVA, which produces the exact same values :)

ranova(model_combined)
```

```
## ANOVA-like table for random-effects: Single term deletions
##
## Model:
## feret_micron ~ method * stage_combined + (1 | specimen_id)
##                    npar logLik   AIC    LRT Df Pr(>Chisq)
## <none>               10 -12612 25243
## (1 | specimen_id)     9 -12946 25910 668.82  1  < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
ranova(model_species_filtered)
```

```
## ANOVA-like table for random-effects: Single term deletions
##
## Model:
## feret_micron ~ method * species + stage + (1 | specimen_id)
##                     npar logLik   AIC    LRT Df Pr(>Chisq)
## <none>                15 -11962 23954
## (1 | specimen_id)     14 -11989 24005 53.453  1  2.648e-13 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
ranova(model_phylum_filtered)
```

```
## ANOVA-like table for random-effects: Single term deletions
##
## Model:
## feret_micron ~ method * phylum + (1 | specimen_id)
```

```
##                    npar logLik   AIC    LRT Df Pr(>Chisq)
## <none>                6 -13068 26148
## (1 | specimen_id)     5 -13115 26239 93.282  1  < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

---

11. SPECIMEN-LEVEL VARIATION

```
# ---- Supp Fig 1 (split by method):  ----

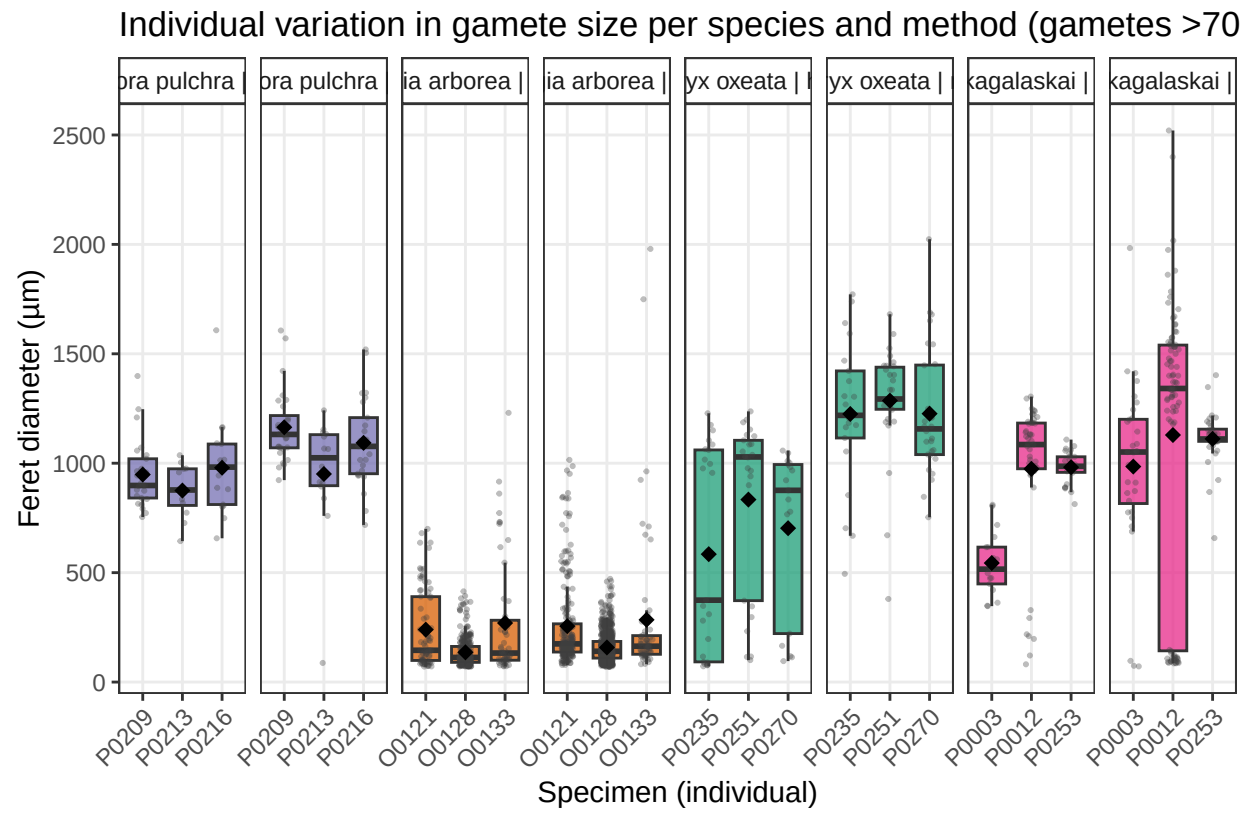
# species_cols stays the same as your original
species_cols <- c(
  "Monanchora pulchra" = "#7570b3",
  "Paragorgia arborea" = "#d95f02",
  "Stelodoryx oxeata"  = "#1b9e77",
  "Tedania kagalaskai" = "#e7298a"
)

# Build the combined facet label with explicit ordering
gametes_large <- gametes_large %>%
  mutate(
    species_method = factor(
      paste(species, method, sep = " | "),
      levels = as.vector(t(outer(
        c("Monanchora pulchra", "Paragorgia arborea",
          "Stelodoryx oxeata",  "Tedania kagalaskai"),
        c("histology", "microCT"),
        paste, sep = " | "
      )))
    )
  )

p_supp1 <- ggplot(gametes_large,
                  aes(x = specimen_id, y = feret_micron, fill = species)) +
  geom_boxplot(outlier.shape = NA, alpha = 0.75, width = 0.7) +
  geom_point(position = position_jitter(width = 0.15),
             size = 0.4, alpha = 0.35, colour = "grey25") +
  stat_summary(fun = mean, geom = "point",
               shape = 18, size = 2.5, colour = "black") +
  facet_wrap(~ species_method, scales = "free_x", nrow = 1) +
  scale_fill_manual(values = species_cols) +
  theme_bw() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    strip.background = element_rect(fill = "white"),
    legend.position = "none",
    panel.grid.minor = element_blank()
  ) +
  labs(
    x = "Specimen (individual)",
```

```
    y = "Feret diameter (µm)",
    title = "Individual variation in gamete size per species and method (gametes >70 µm)",
    caption = "Panels run left to right: each species is shown for histology then microCT. Grey dots = i
  )

p_supp1
```



ecies is shown for histology then microCT. Grey dots = individual measurements, black diamonds = individual means.

```
# ---- Supp Fig 2 (split by method):

# 12 specimen colours (same as your original)
specimen_cols <- c(
  "P0209" = "#1b9e77", "P0213" = "#66c2a5", "P0216" = "#006d2c",
  "O0121" = "#d95f02", "O0128" = "#e7298a", "O0133" = "#fc8d62",
  "P0235" = "#7570b3", "P0251" = "#984ea3", "P0270" = "#bcbddc",
  "P0003" = "#e6ab02", "P0012" = "#a6761d", "P0253" = "#ffd92f"
)

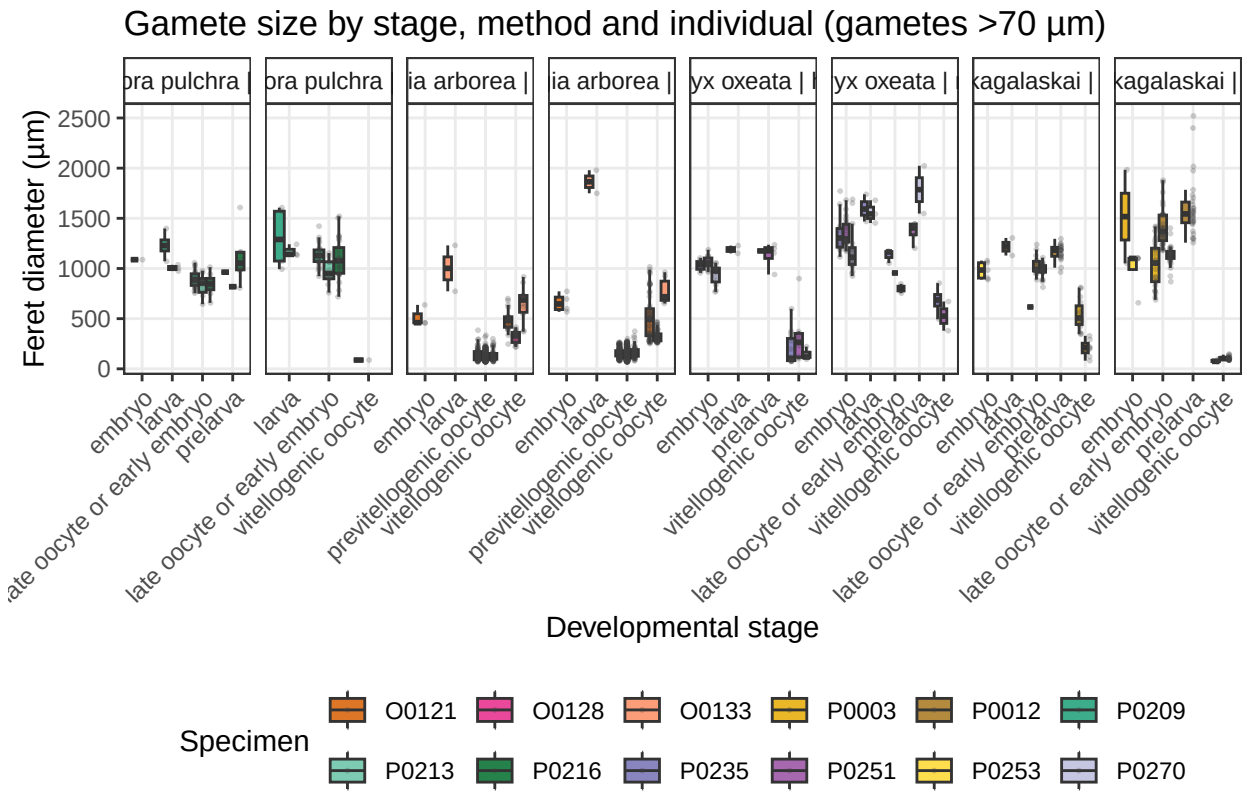
# Build the combined facet label with explicit ordering
gametes_large <- gametes_large %>%
  mutate(
    species_method = factor(
      paste(species, method, sep = " | "),
      levels = as.vector(t(outer(
        c("Monanchora pulchra", "Paragorgia arborea",
          "Stelodoryx oxeata",  "Tedania kagalaskai"),
```

```
        c("histology", "microCT"),
        paste, sep = " | "
      )))
    )
  )

# Set specimen order to match the specimen colour palette
specimen_levels <- names(specimen_cols)

p_supp2 <- ggplot(gametes_large,
                  aes(x = stage, y = feret_micron, fill = specimen_id)) +
  geom_boxplot(outlier.shape = NA, alpha = 0.85,
               position = position_dodge(preserve = "single")) +
  geom_point(position = position_jitterdodge(jitter.width = 0.08,
                                             dodge.width = 0.75),
             size = 0.35, alpha = 0.25, colour = "grey25") +
  facet_wrap(~ species_method, scales = "free_x", nrow = 1) +
  scale_fill_manual(values = specimen_cols, name = "Specimen") +
  theme_bw() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    strip.background = element_rect(fill = "white"),
    legend.position = "bottom",
    panel.grid.minor = element_blank()
  ) +
  guides(fill = guide_legend(nrow = 2, byrow = TRUE)) +
  labs(
    x = "Developmental stage",
    y = "Feret diameter (µm)",
    title = "Gamete size by stage, method and individual (gametes >70 µm)",
    caption = "Panels run left to right: each species is shown for histology then microCT. Each boxplot
  )

p_supp2
```



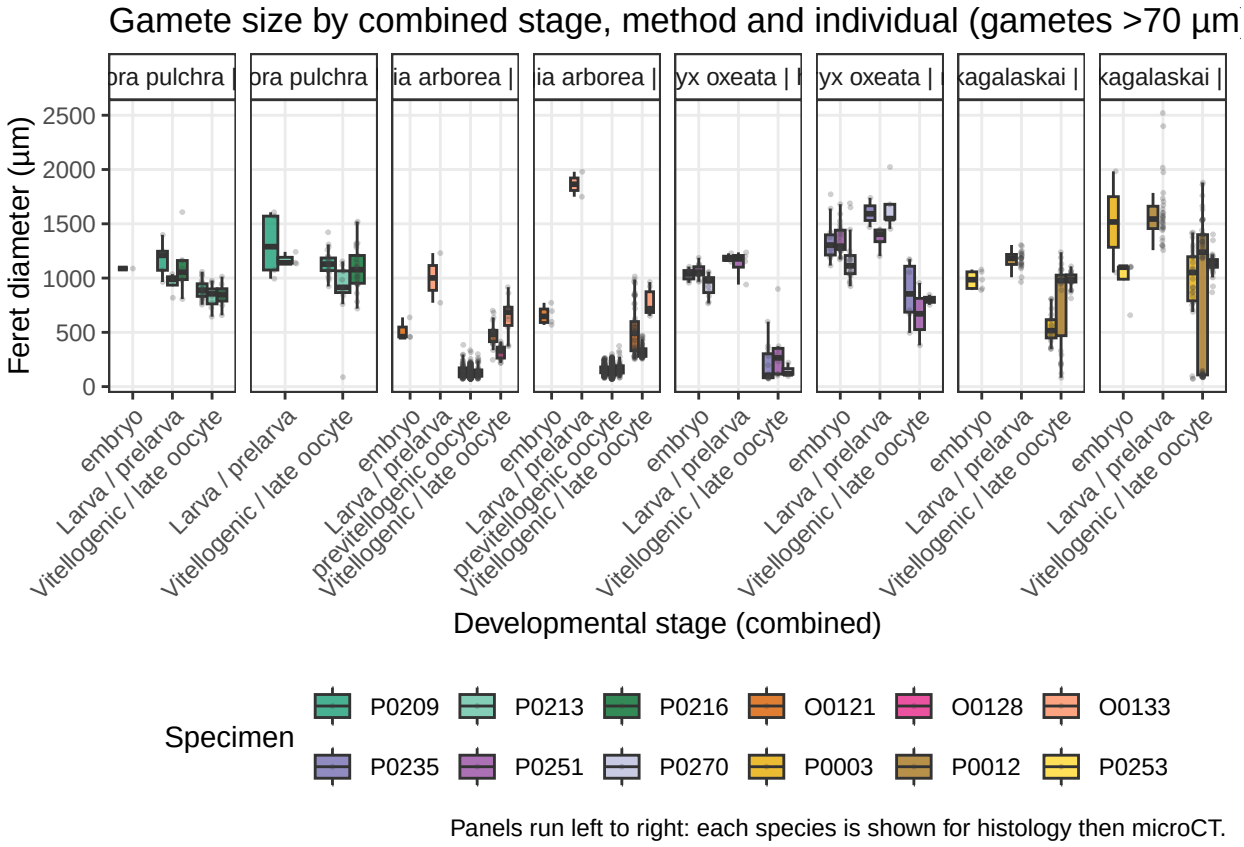
hown for histology then microCT. Each boxplot = one individual; colours distinguish the three specimens per species.

```r
# Make sure gametes_filtered has the same factor levels / column
gametes_filtered <- gametes_filtered %>%
  mutate(
    specimen_id = factor(specimen_id, levels = specimen_levels),
    species     = factor(species, levels = c("Monanchora pulchra",
                                             "Paragorgia arborea",
                                             "Stelodoryx oxeata",
                                             "Tedania kagalaskai")),
    species_method = factor(
      paste(species, method, sep = " | "),
      levels = as.vector(t(outer(
        c("Monanchora pulchra", "Paragorgia arborea",
          "Stelodoryx oxeata",  "Tedania kagalaskai"),
        c("histology", "microCT"),
        paste, sep = " | "
      )))
    )
  )

p_supp2_combined <- ggplot(gametes_filtered,
                           aes(x = stage_combined, y = feret_micron,
                               fill = specimen_id)) +
  geom_boxplot(outlier.shape = NA, alpha = 0.8,
               position = position_dodge(preserve = "single")) +
  geom_point(position = position_jitterdodge(jitter.width = 0.08,
                                             dodge.width = 0.75),
```

```
            size = 0.35, alpha = 0.25, colour = "grey25") +
  facet_wrap(~ species_method, scales = "free_x", nrow = 1) +
  scale_fill_manual(values = specimen_cols, name = "Specimen") +
  theme_bw() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    strip.background = element_rect(fill = "white"),
    legend.position = "bottom",
    panel.grid.minor = element_blank()
  ) +
  guides(fill = guide_legend(nrow = 2, byrow = TRUE)) +
  labs(
    x = "Developmental stage (combined)",
    y = "Feret diameter (µm)",
    title = "Gamete size by combined stage, method and individual (gametes >70 µm)",
    caption = "Panels run left to right: each species is shown for histology then microCT."
  )

p_supp2_combined
```



(ICC) - maybe not needed, since I have too few specimens

```
#Using ICC = Intraclass Correlation Coefficient Analysis
#Can say how much of the total variation in gamete size is due to differences between specimens (as opp
# So 0 = no variation, all specimens are identical
# If ICC > 0.5 then high, specimens are very different from eachother
```

```r
# ---- 11a. ICC for filtered model ----
icc(m2) # here, for example adjusted ICC is 0.09, and unadjusted is 0.018 - after accounting for method
```

```
## # Intraclass Correlation Coefficient
##
##     Adjusted ICC: 0.094
##   Unadjusted ICC: 0.018
```

```r
# Unadjusted ICC = the proportion of variance explained by the random effect without accounting for fix


# ---- 11b. Phylum-specific ICC ----
icc_porifera <- performance::icc(
  lmer(feret_micron ~ method + stage + (1 | specimen_id),
       data = gametes %>% filter(phylum == "Porifera"))
)
print("ICC for Porifera:")
```

```
## [1] "ICC for Porifera:"
```

```r
print(icc_porifera) # here: In sponges, 16.1% of the variation is between specimens. but here, we also
```

```
## # Intraclass Correlation Coefficient
##
##     Adjusted ICC: 0.161
##   Unadjusted ICC: 0.030
```

```r
icc_cnidaria <- performance::icc(
  lmer(feret_micron ~ method + stage + (1 | specimen_id),
       data = gametes %>% filter(phylum == "Cnidaria"))
)
print("ICC for Cnidaria:")
```

```
## [1] "ICC for Cnidaria:"
```

```r
print(icc_cnidaria)
```

```
## # Intraclass Correlation Coefficient
##
##     Adjusted ICC: 0.108
##   Unadjusted ICC: 0.038
```

```r
# ---- 11c. Overall ICC (all data) ----
icc_overall <- performance::icc(
  lmer(feret_micron ~ method + stage + (1 | specimen_id),
       data = gametes)
)
print("ICC for all data:")
```

```
## [1] "ICC for all data:"
```

```r
print(icc_overall)
```

```
## # Intraclass Correlation Coefficient
##
##     Adjusted ICC: 0.418
##   Unadjusted ICC: 0.054
```

```r
#Interpretation: Within each phylum, specimens are quite similar (low ICC). Between phyla, there is a l
```

---

12. OVERALL BIAS SUMMARY

```r
# Using filtered paired data
paired_filtered %>%
  summarise(
    mean_bias = mean(diff, na.rm = TRUE),
    sd_bias = sd(diff, na.rm = TRUE),
    n = n()
  )
```

```
## # A tibble: 1 x 3
##   mean_bias sd_bias     n
##       <dbl>   <dbl> <int>
## 1      200.    228.    27
```

```r
# Overall percentage bias
mean(paired_filtered$diff / paired_filtered$histology, na.rm = TRUE) * 100
```

```
## [1] 35.28864
```

```r
# Stage-specific bias
paired_filtered %>%
  group_by(stage) %>%
  summarise(
    mean_bias = mean(diff, na.rm = TRUE),
    percent_bias = mean((microCT - histology) / histology, na.rm = TRUE) * 100
  )
```

```
## # A tibble: 6 x 3
##   stage                        mean_bias percent_bias
##   <chr>                            <dbl>        <dbl>
## 1 embryo                           196.          22.0
## 2 larva                            379.          35.9
## 3 late oocyte or early embryo      265.          32.6
## 4 prelarva                         338.          29.4
## 5 previtellogenic oocyte            42.2         37.4
## 6 vitellogenic oocyte               71.4         47.5
```

```r
gametes_filtered %>% dplyr::count(stage_combined, method)
```

```
##               stage_combined    method   n
## 1           Larva / prelarva histology  49
## 2           Larva / prelarva   microCT  53
## 3 Vitellogenic / late oocyte histology 174
## 4 Vitellogenic / late oocyte   microCT 307
## 5                     embryo histology  42
## 6                     embryo   microCT  64
## 7     previtellogenic oocyte histology 358
## 8     previtellogenic oocyte   microCT 857
```

```r
emm_phylum <- emmeans(model_phylum_filtered, ~ method | phylum)
contrast(emm_phylum, "pairwise", by = "phylum", adjust = "tukey")
```

```
## phylum = Cnidaria:
##  contrast            estimate   SE   df t.ratio p.value
##  histology - microCT      -20 13.7 1892  -1.455  0.1458
##
## phylum = Porifera:
##  contrast            estimate   SE   df t.ratio p.value
##  histology - microCT     -287 21.0 1895 -13.687 <0.0001
##
## Degrees-of-freedom method: kenward-roger
```

---

13. FECUNDITY (SPONGES)

```r
# Using microCT data only
microct_data <- data %>% filter(method == "microCT")

gamete_counts <- microct_data %>%
  filter(grepl("Gamete", type, ignore.case = TRUE)) %>%
  group_by(specimen_id) %>%
  summarise(n_gametes = n(), .groups = "drop")

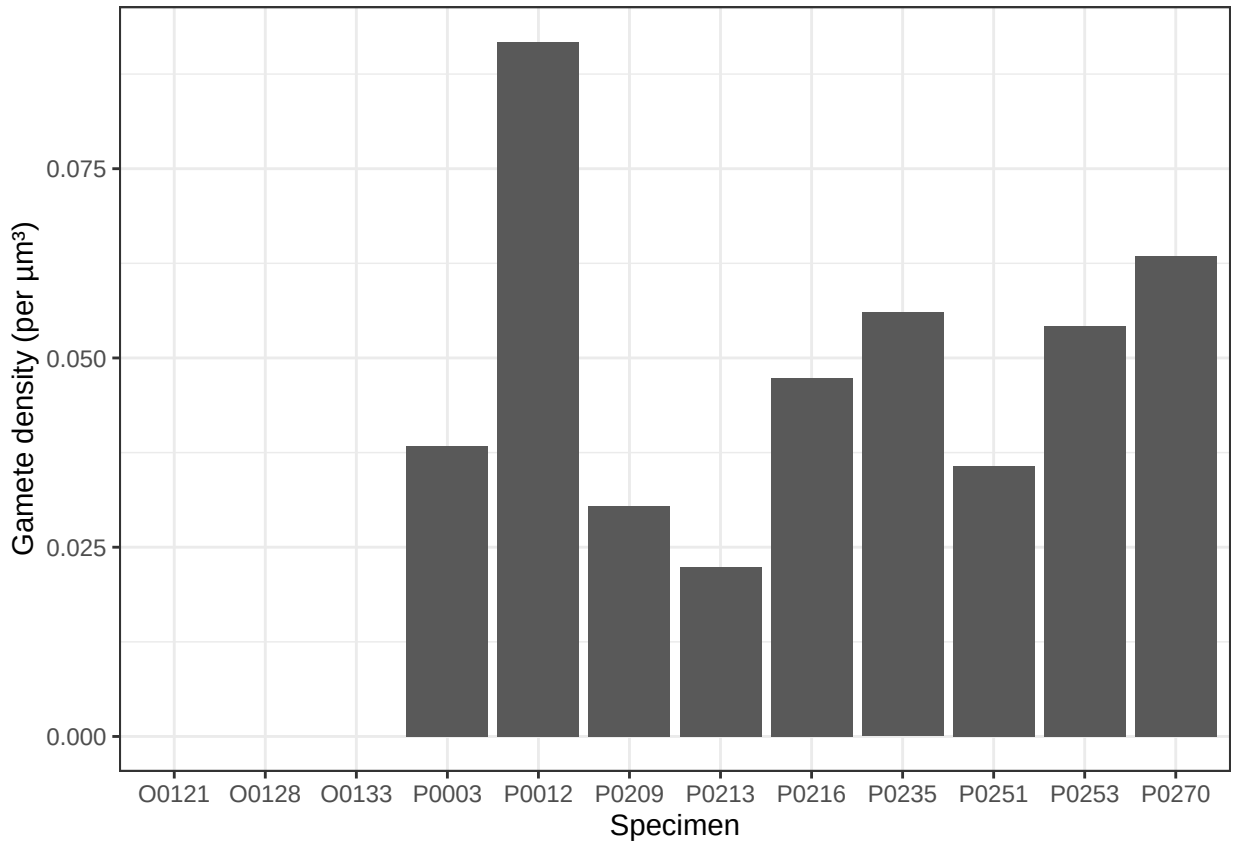
tissue_volume <- microct_data %>%
  filter(type %in% c("Mesohyl", "Spicules")) %>%
  group_by(specimen_id) %>%
  summarise(tissue_volume = sum(volume, na.rm = TRUE), .groups = "drop")

fecundity <- gamete_counts %>%
  left_join(tissue_volume, by = "specimen_id") %>%
  mutate(
    gamete_density = n_gametes / tissue_volume,
    gamete_density_log = log(gamete_density)
  )

print(fecundity)
```

```
## # A tibble: 12 x 5
##    specimen_id n_gametes tissue_volume gamete_density gamete_density_log
##    <chr>           <int>         <dbl>          <dbl>               <dbl>
##  1 O0121             222            NA         NA                  NA
##  2 O0128             708            NA         NA                  NA
##  3 O0133              60            NA         NA                  NA
##  4 P0003              32          833.          0.0384             -3.26
##  5 P0012              86          937.          0.0918             -2.39
##  6 P0209              28          919.          0.0305             -3.49
##  7 P0213              14          626.          0.0224             -3.80
##  8 P0216              25          528.          0.0474             -3.05
##  9 P0235              21          375.          0.0560             -2.88
## 10 P0251              30          839.          0.0357             -3.33
## 11 P0253              33          608.          0.0543             -2.91
## 12 P0270              28          441.          0.0634             -2.76
```

```r
# Optional fecundity plot (commented out)
ggplot(fecundity, aes(x = specimen_id, y = gamete_density)) +
geom_col() + labs(y = "Gamete density (per µm³)", x = "Specimen") + theme_bw()
```



14. PARAGORGIA SIZE DISTRIBUTION (ADDITIONAL MUSEUM DATA)

```
# Load museum specimen data (additional microCT measurements)
museum <- read.csv("Paragorgia_microCT_museum.csv")

# Clean museum column names (if not already clean)
names(museum) <- c(
  "specimen_id", "species_replicate", "phylum", "species", "stage",
  "type", "measurement", "volume", "surface_area_lindblad_2005_mm2",
  "aspect_ratio", "feret_min_mm", "feret_mean_mm", "feret_max_mm",
  "feret_micron", "notes", "annotation_method", "method"
)

# Subset Paragorgia microCT data from main dataset
paragorgia_microCT <- gametes[
  gametes$species == "Paragorgia arborea" &
  gametes$method == "microCT",
]

# Combine with museum data
paragorgia_all <- rbind(paragorgia_microCT, museum)

# Plot: density with stage-coloured rug
ggplot(paragorgia_all, aes(x = feret_micron)) +

  # Developmental stage zones (background shading)
  annotate("rect", xmin = 0, xmax = 400,  ymin = -Inf, ymax = Inf,
           fill = "#1b9e77", alpha = 0.05) +
  annotate("rect", xmin = 400, xmax = 900, ymin = -Inf, ymax = Inf,
           fill = "#d95f02", alpha = 0.05) +
  annotate("rect", xmin = 900, xmax = 2000, ymin = -Inf, ymax = Inf,
           fill = "#7570b3", alpha = 0.05) +

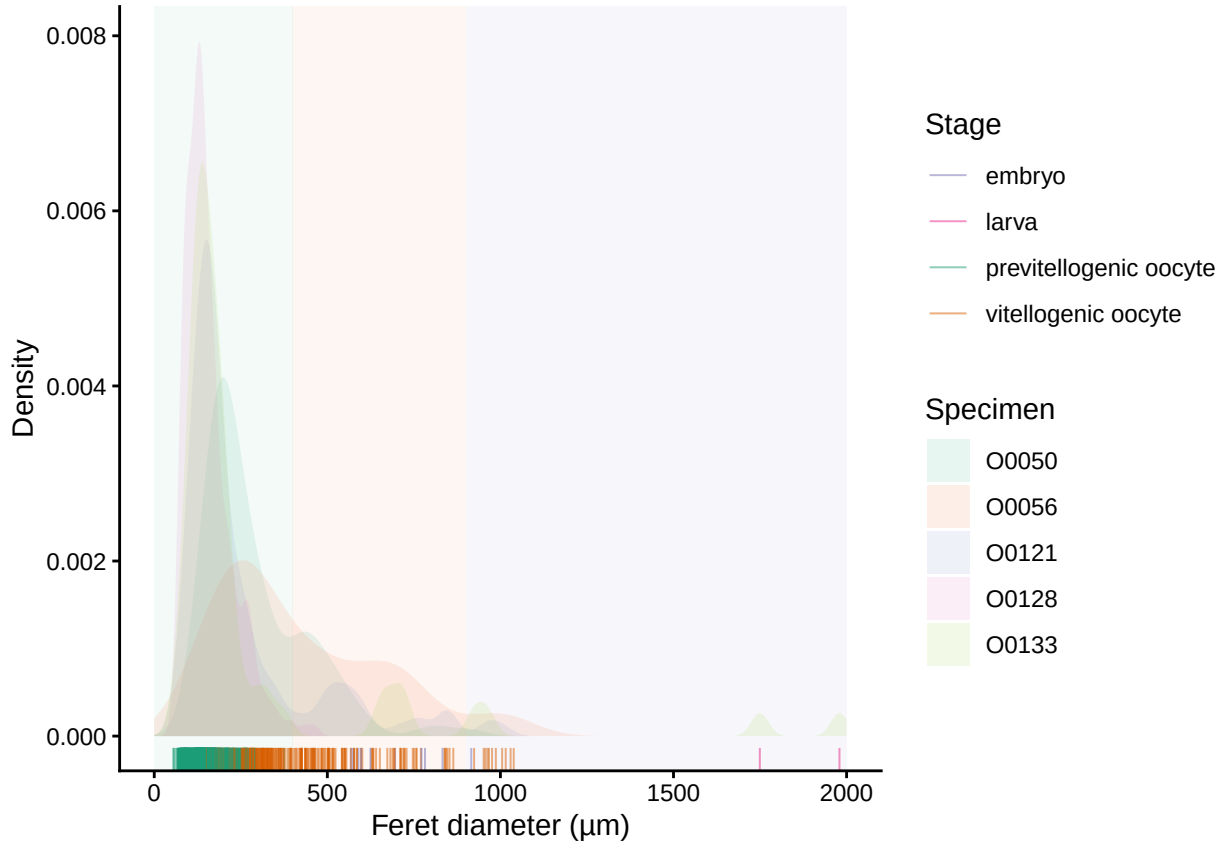
  # Kernel density estimates (coloured by specimen)
  geom_density(aes(fill = specimen_id), alpha = 0.15, color = NA) +

  # Rug plot (individual observations coloured by stage)
  geom_rug(aes(color = stage), sides = "b", alpha = 0.5, linewidth = 0.35) +

  scale_fill_brewer(palette = "Set2") +
  scale_color_manual(values = c(
    "previtellogenic oocyte" = "#1b9e77",
    "vitellogenic oocyte"    = "#d95f02",
    "embryo"                 = "#7570b3",
    "larva"                  = "#e7298a",
    "prelarva"               = "#66a61e"
  )) +

  theme_classic() +
  labs(
    x = "Feret diameter (µm)",
    y = "Density",
    fill = "Specimen",
```

```
    color = "Stage"
  )
```




```
# ---- Summary statistics for Paragorgia stages ----
paragorgia_stats <- gametes %>%
  filter(species == "Paragorgia arborea") %>%
  group_by(method, stage) %>%
  summarise(
    n = n(),
    mean_feret = mean(feret_micron, na.rm = TRUE),
    sd_feret = sd(feret_micron, na.rm = TRUE),
    min_feret = min(feret_micron, na.rm = TRUE),
    max_feret = max(feret_micron, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(stage, method)

print(paragorgia_stats)
```

```
## # A tibble: 8 x 7
##   method    stage                    n mean_feret sd_feret min_feret max_feret
##   <chr>     <chr>                <int>      <dbl>    <dbl>     <dbl>     <dbl>
## 1 histology embryo                   3       518.    104.      457.       637.
## 2 microCT   embryo                   4       658.     93.2     568.       772.
## 3 histology larva                    2      1001.    324.      772.      1230.
```

```
## 4 microCT   larva                      2      1864.    163.     1749.      1979.
## 5 histology previtellogenic oocyte   454       111.     54.9      24.4      385.
## 6 microCT   previtellogenic oocyte   863       146.     48.5      54.9      375.
## 7 histology vitellogenic oocyte       47       456.    170.      218.       916.
## 8 microCT   vitellogenic oocyte      121       431.    197.      251.      1015.
```

---

15. SESSION INFO

```
sessionInfo()
```

```
## R version 4.4.1 (2024-06-14)
## Platform: aarch64-apple-darwin20
## Running under: macOS 27.0
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib;  LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Oslo
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] RLRsim_3.1-9       see_0.13.0         performance_0.16.0 emmeans_2.0.4
##  [5] lmerTest_3.2-1     lme4_2.0-6         Matrix_1.7-5       ggplot2_4.0.3
##  [9] tidyr_1.3.2        dplyr_1.2.1        janitor_2.2.1
##
## loaded via a namespace (and not attached):
##  [1] gtable_0.3.6        bayestestR_0.18.1   xfun_0.60
##  [4] ggrepel_0.9.8       insight_1.7.0       lattice_0.22-9
##  [7] numDeriv_2016.8-1.1 vctrs_0.7.3         tools_4.4.1
## [10] Rdpack_2.6.6        generics_0.1.4      datawizard_1.3.1
## [13] parallel_4.4.1      pbkrtest_0.5.5      sandwich_3.1-3
## [16] tibble_3.3.1        pkgconfig_2.0.3     RColorBrewer_1.1-3
## [19] S7_0.2.2            lifecycle_1.0.5     compiler_4.4.1
## [22] farver_2.1.2        stringr_1.6.0       codetools_0.2-20
## [25] snakecase_0.11.1    htmltools_0.5.9     yaml_2.3.12
## [28] pillar_1.11.1       nloptr_2.2.1        MASS_7.3-66
## [31] reformulas_0.4.4    boot_1.3-32         multcomp_1.4-31
## [34] nlme_3.1-168        tidyselect_1.2.1    digest_0.6.39
## [37] mvtnorm_1.3-6       stringi_1.8.9       purrr_1.2.2
## [40] labeling_0.4.3      splines_4.4.1       fastmap_1.2.0
## [43] grid_4.4.1          cli_3.6.6           magrittr_2.0.5
## [46] patchwork_1.3.2     survival_3.8-9      utf8_1.2.6
## [49] broom_1.0.13        TH.data_1.1-5       withr_3.0.3
```

```
## [52] backports_1.5.1     scales_1.4.0        lubridate_1.9.5
## [55] timechange_0.4.0    estimability_2.0.0  rmarkdown_2.31
## [58] otel_0.2.0          zoo_1.9-0           evaluate_1.0.5
## [61] knitr_1.51          parameters_0.29.2   rbibutils_2.4.1
## [64] mgcv_1.9-4          rlang_1.3.0         Rcpp_1.1.2
## [67] xtable_1.8-8        glue_1.8.1          rstudioapi_0.19.0
## [70] minqa_1.2.8         R6_2.6.1
```